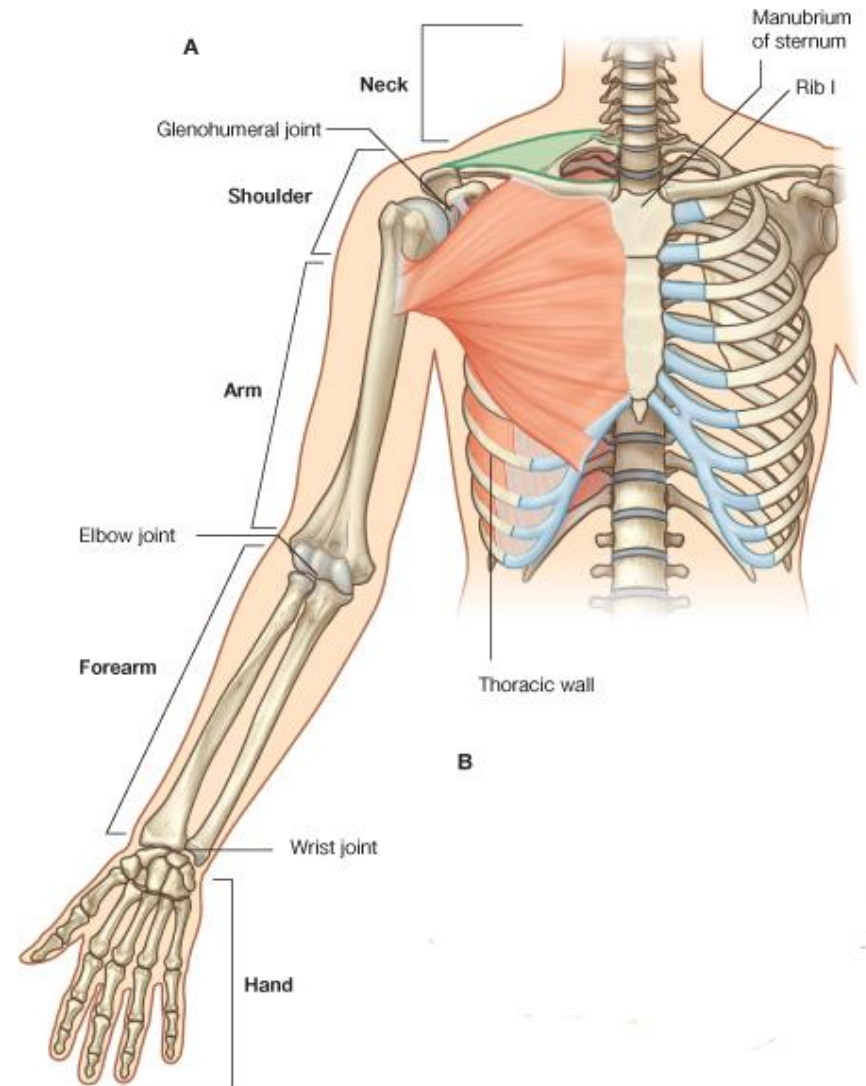


Upper Extremity: Overview/Review

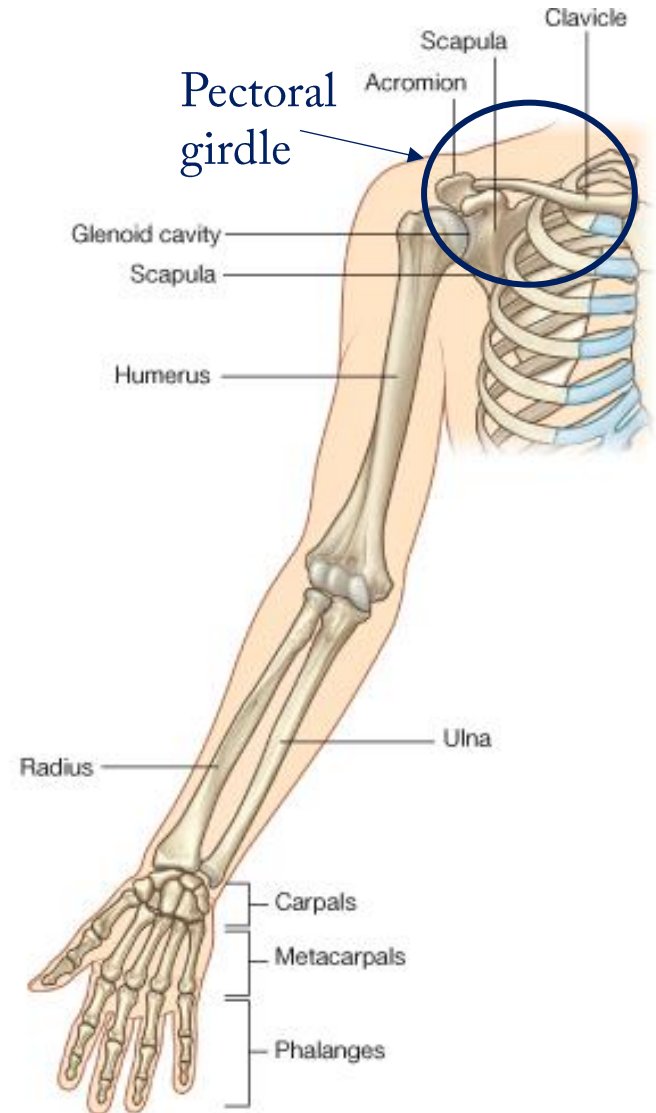


IS-MSN 2024
Mark Teaford, Ph.D.
Andrea Taylor, Ph.D.
Theo Smith, Ph.D.
Amanda Smith, Ph.D.

- The upper extremity largely comprises the shoulder, arm (brachium), forearm (antebrachium), and hand.
- The upper extremity also has connections with parts of the neck and thoracic wall.
- The upper limb is highly mobile. The system is designed to facilitate functional reach and positioning and use of the hand.



- The bones of the upper extremity include:
 - Clavicle
 - Scapula
 - Humerus
 - Radius
 - Ulna
 - Carpals
 - Metacarpals
 - Phalanges
- The **pectoral girdle** comprises the scapula and clavicle and serves to connect the upper limb to the axial skeleton.



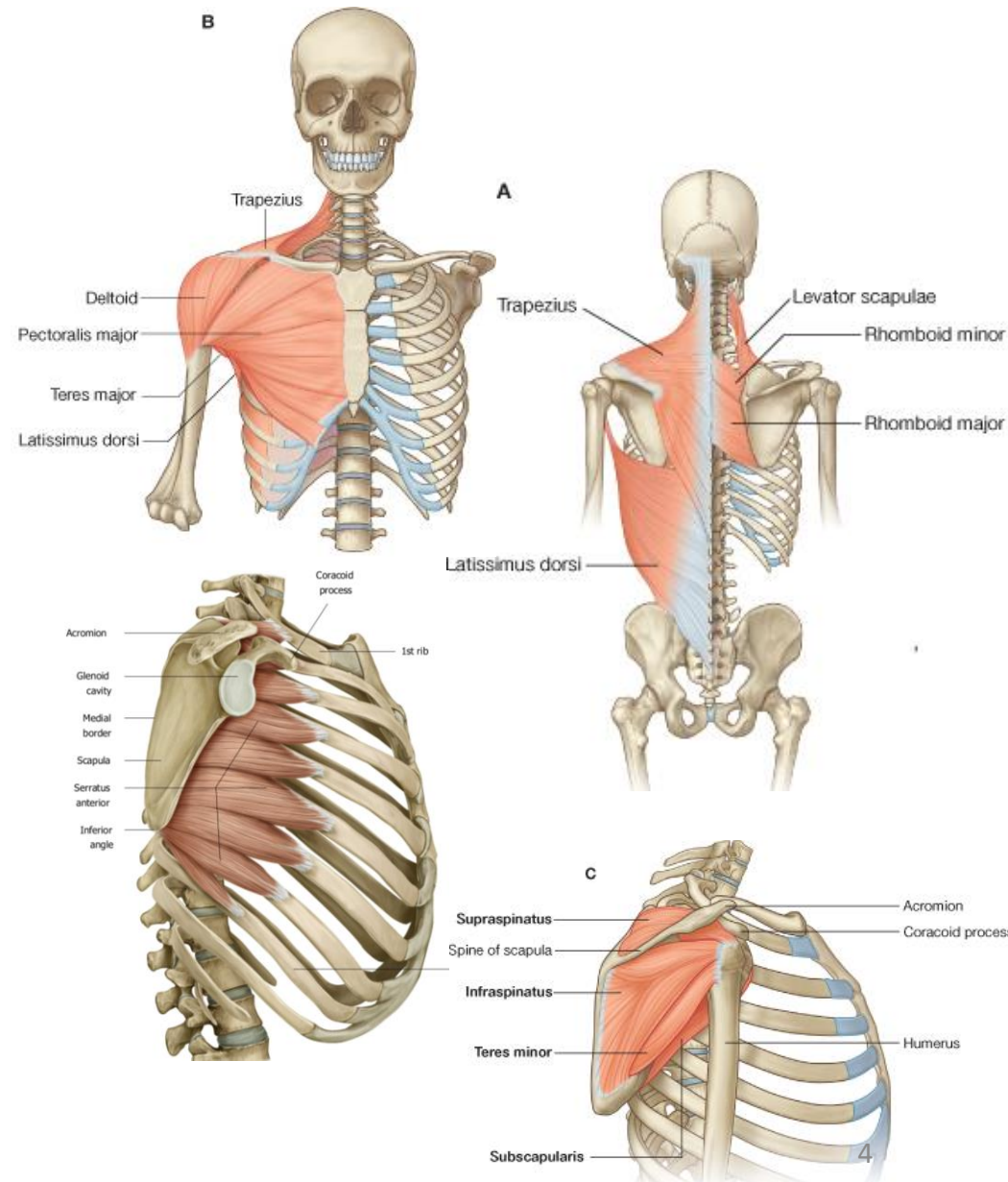
Organization of muscle compartments: Shoulder

Muscles that connect the scapula and clavicle to the trunk:

- Trapezius
- Levator scapulae
- Rhomboids (major and minor)
- Subclavius
- Serratus anterior

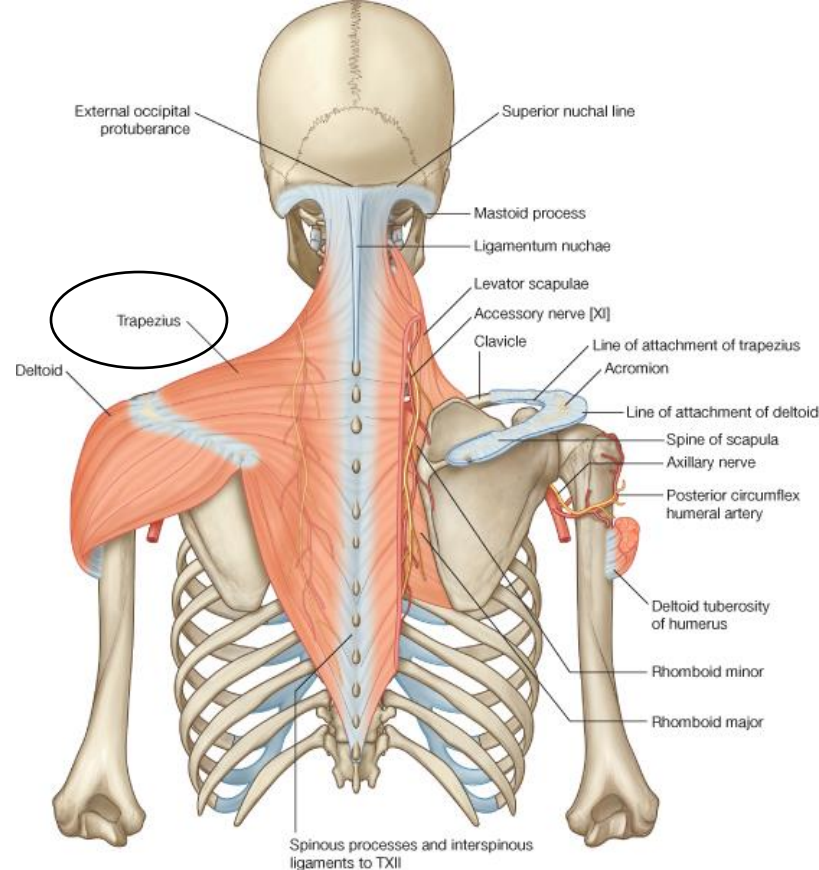
Muscles that connect the clavicle, scapula and body wall to the proximal humerus:

- Pectoral muscles (major and minor)
- Latissimus dorsi
- Teres major
- Deltoid
- Rotator cuff muscles: supraspinatus, infraspinatus, teres minor, subscapularis
- All of these muscles except trapezius are *hypaxial* muscles that migrate dorsally during development. As hypaxial muscles, they are all innervated by ventral rami, specifically, nerves of the brachial plexus.
- The trapezius has a different embryologic development and thus a different source of innervation (*spinal accessory n.* – or *cranial nerve XI*).

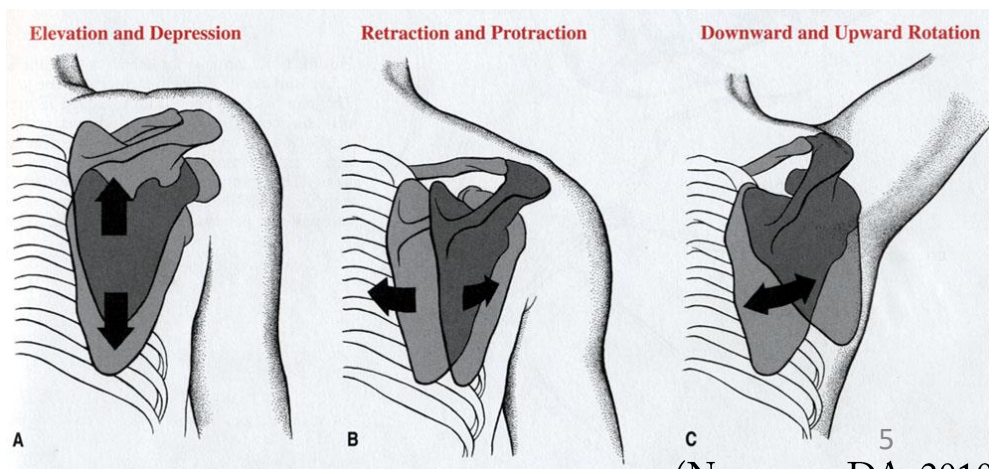


Trapezius muscle

- Extensive origin from:
 - Superior nuchal line
 - External occipital protuberance
 - Ligamentum nuchae (Nuchal ligament)
 - C7-T12 spinous processes
- Insertion:
 - Lateral 1/3rd of clavicle, acromion & scapular spine
- Actions:
 - Upper fibers **elevate** scapula
 - Lower fibers **depress** scapula
 - Upper & lower fibers together **upwardly rotate** scapula (along with serratus anterior)
 - Middle fibers **retract** scapula



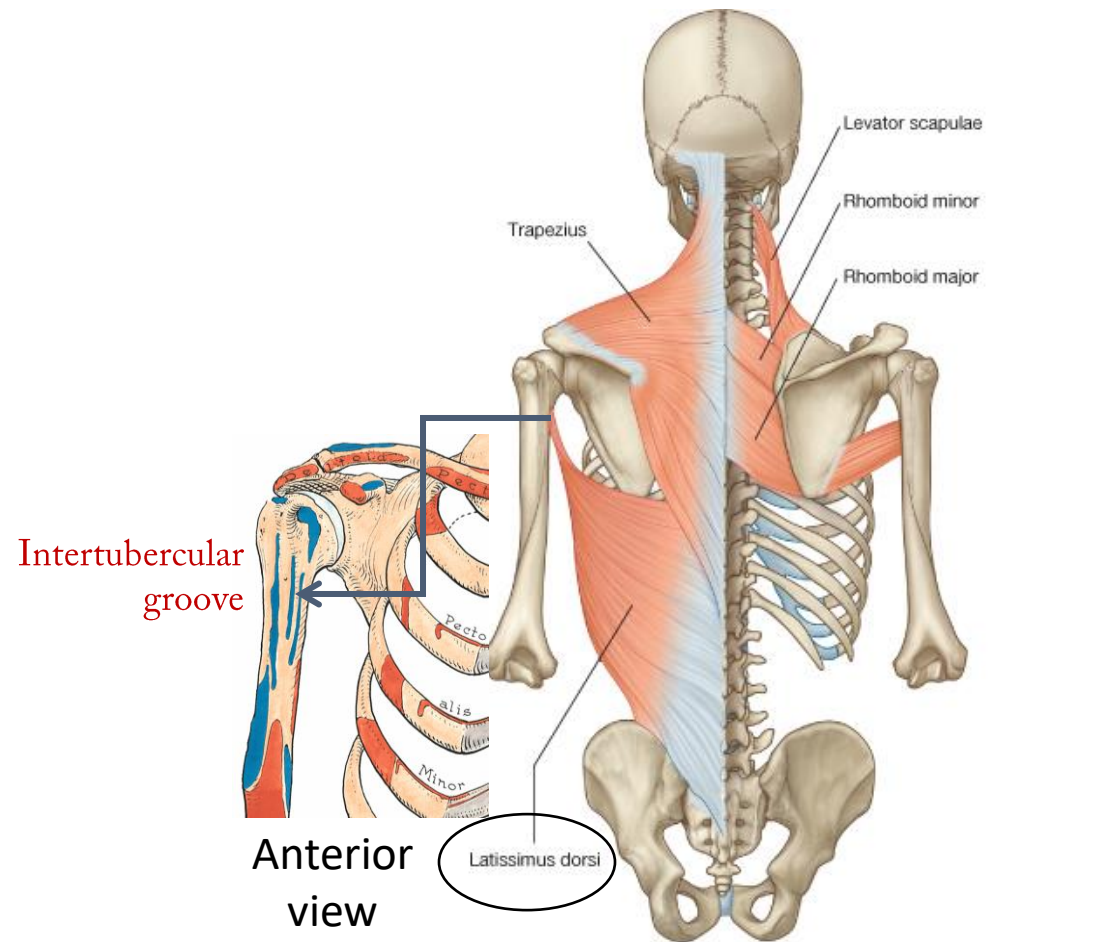
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Latissimus dorsi

- Extensive origin from spinous processes T7-S5, ribs 9-12, thoracolumbar fascia and iliac crest (note the overlap by the trapezius muscle)
- Insertion into the **intertubercular groove** of humerus
- Actions:
 - Extends, internally (medially) rotates, and adducts humerus.
 - *Alternative action:* With the humeral attachment fixed at the intertubercular groove, the latissimus dorsi can raise the body towards the upper trunk, as in hoisting the body up on parallel bars.
- Innervation: thoracodorsal nerve (C6-C8)

Posterior view

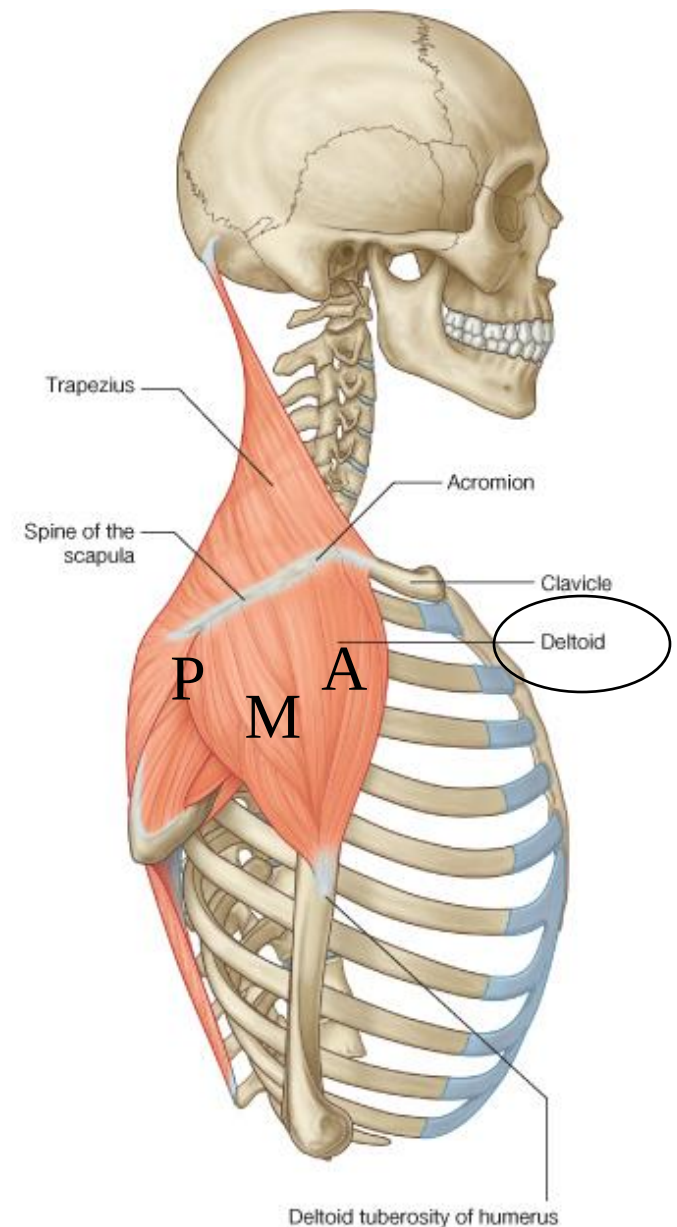


Anterior view

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Deltoid

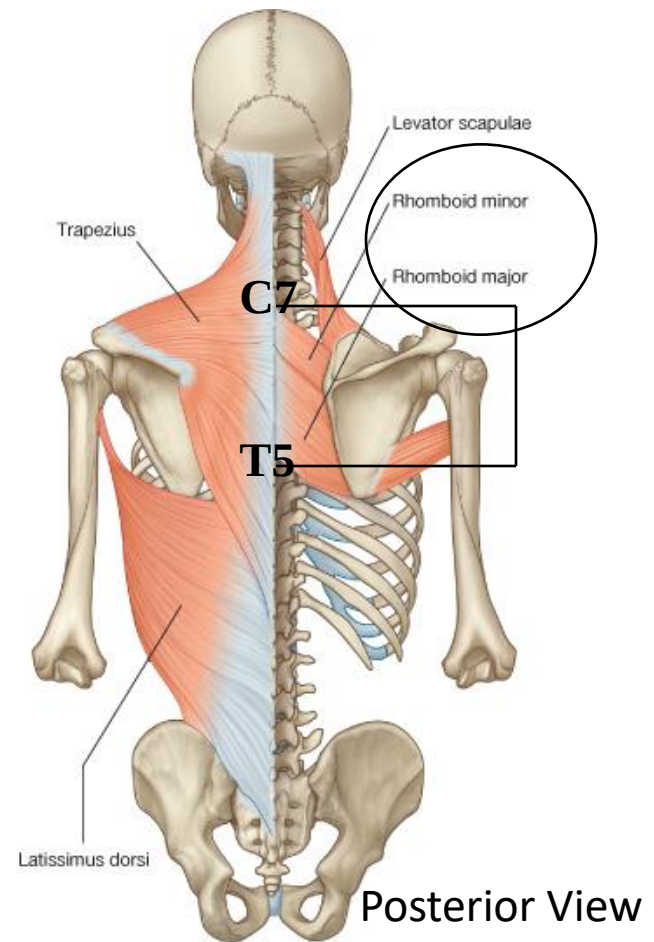
- Originates from the lateral 1/3rd of clavicle, acromion, and scapular spine
- Comprises three separate heads – Anterior (A), Middle (M), Posterior (P) – inserts as convergent tendon into the **deltoid tuberosity** of humerus
- **Actions:**
 - Chief humeral abductor (following initiation of abduction by the supraspinatus muscle)
 - Anterior deltoid (A): chief humeral flexor & medially rotates humerus
 - Posterior head (P): extends & laterally rotates humerus
 - Middle head (M): primary abductor of humerus in coronal/frontal plane
- **Innervation:** Axillary nerve (C5-C6)



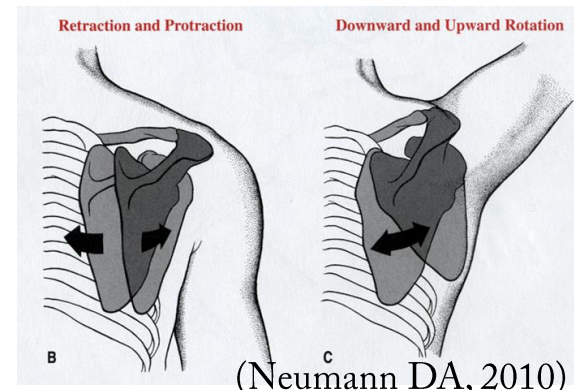
Rhomboid major and minor

- Rhomboid major originates from spinous processes T2-T5
- Rhomboid minor originates from spinous processes C7-T1
- Insert together along the medial border of scapula from scapular spine to inferior angle
- **Actions:**
 - Scapular retraction (brings the scapula closer to the vertebral column (part of the axial skeleton))
 - Downward rotation of the scapula (rotates the glenoid fossa of the scapula inferiorly)
- **Innervation:** dorsal scapular nerve (C5), plus C3-C4 segments

❖ Although these muscles have separate origins, they are often considered to be a single muscle as they are functionally recruited at the same time for a given action and have the same innervation (see slide 20).



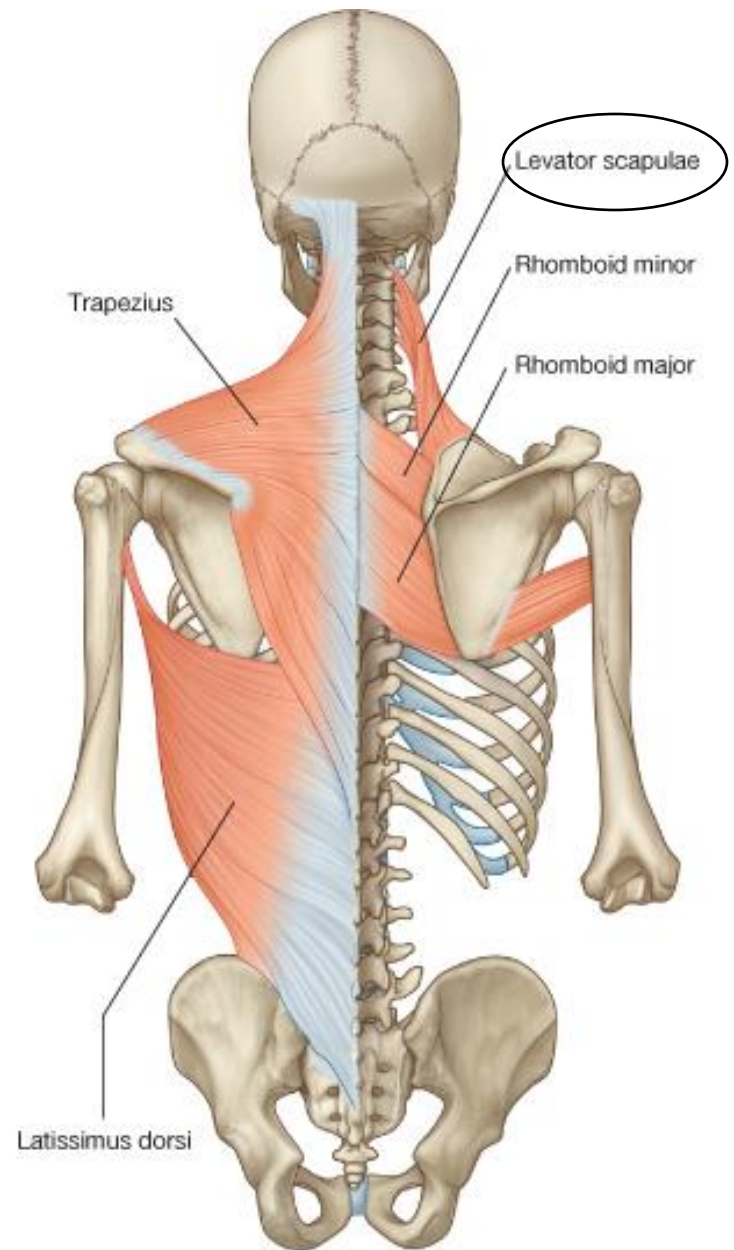
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(Neumann DA, 2010)

Levator scapulae

- **Origin** from transverse processes C1-C4
- **Inserts** into superior angle and upper medial border of scapula
- **Actions:**
 - Scapular elevation
 - Downward rotation of scapula
 - Reverse actions: With scapula fixed, lateral flexion of cervical spine to the same side (**ipsilateral side**) as contraction; bilaterally assists in extension of the cervical spine.
- **Innervation:** dorsal scapular nerve (C5), plus C3-C4 segments



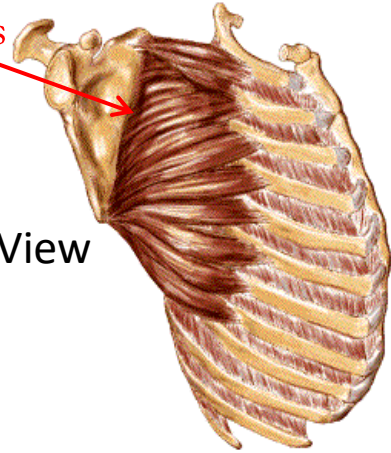
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Serratus anterior

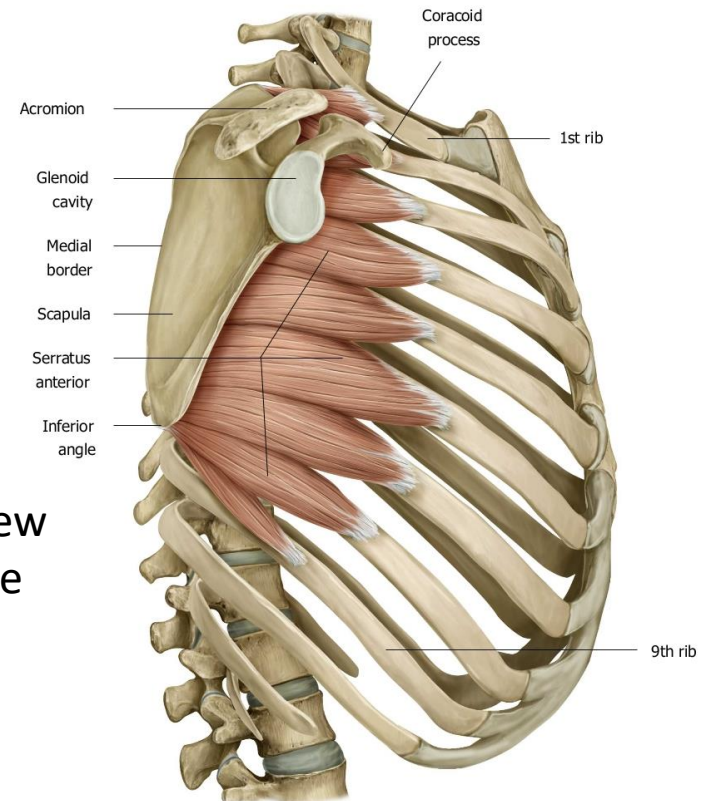
- Originates as fleshy segments attaching to the lateral aspects of the upper 8-9 ribs and deep fascia of intercostal spaces.
- Inserts onto the anterior (ventral) surface of the medial border of the scapula.
- Innervation: long thoracic nerve (C5-C7)

Attaches to the ventral medial border of the scapula (Note: does NOT have an attachment to the subscapular fossa)

Anterolateral View

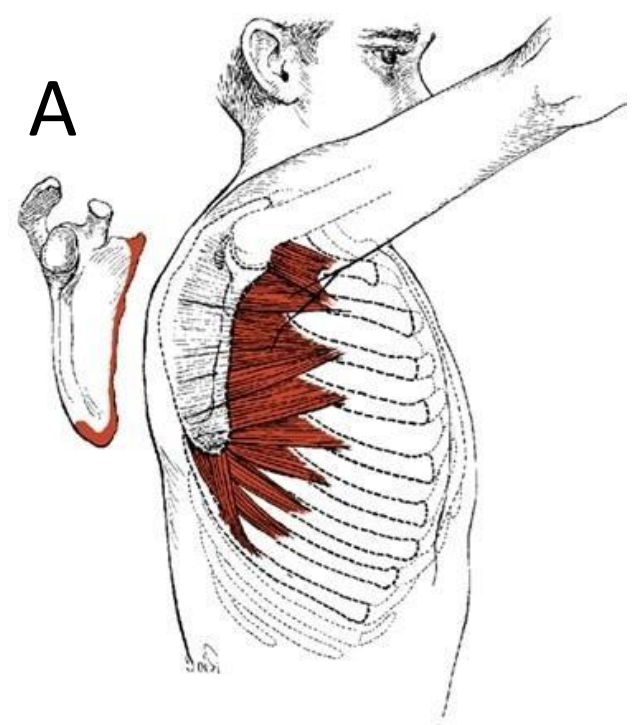


Lateral View
Right side

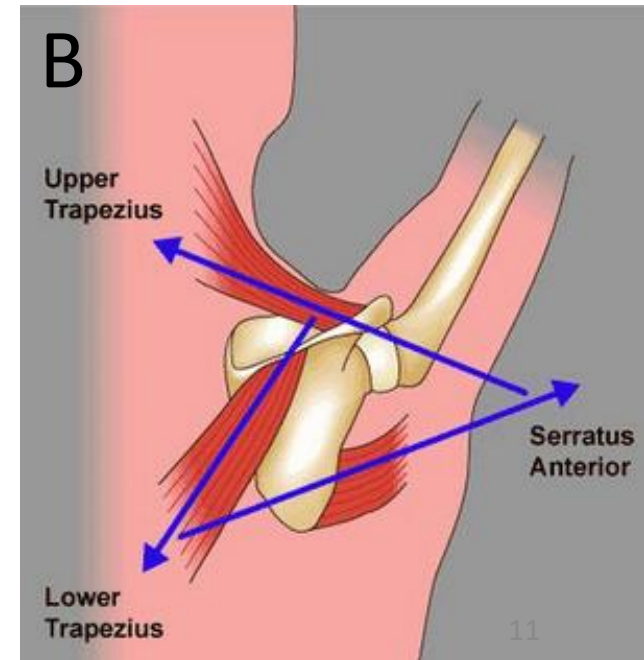


Serratus anterior has two critical functions:

1. **Scapular protraction (Fig. A).** Proper positioning of the scapula on the thoracic wall is important for maintaining scapular stability and promoting normal movements of the upper limb. Scapular protraction occurs when the muscle fibers shorten and the serratus fibers that attach to the ribs are “fixed”, and when the muscle contracts, the fibers shorten and bring the ventral-medial border of the scapula closer to the thoracic wall. This action occurs, for example, when you punch or push, as in the figure at the top.

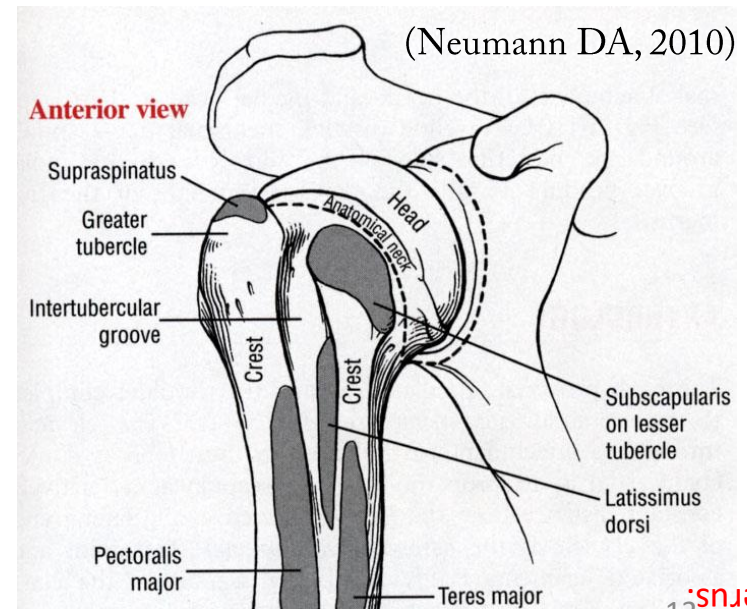
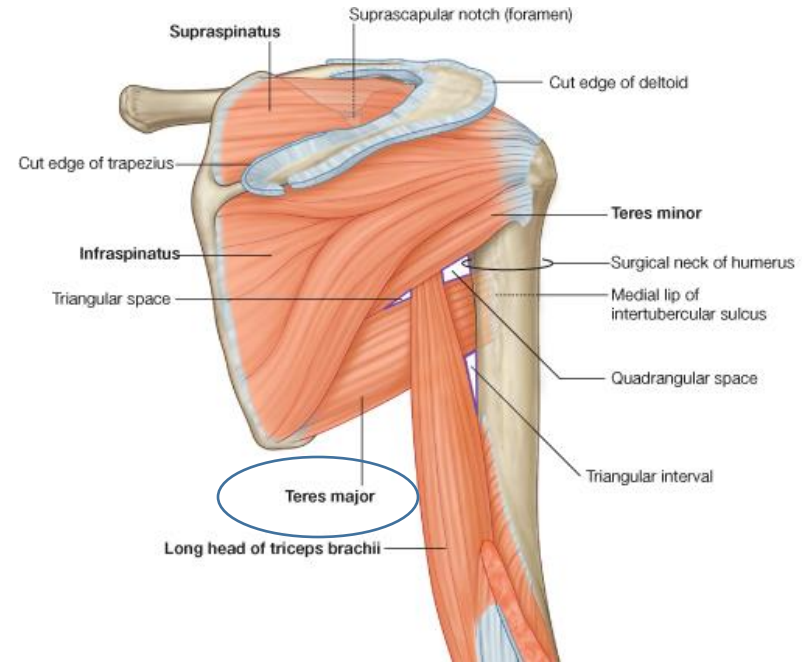


2. **Upward rotation of the scapula (Fig. B).** Upward rotation of the scapula is crucial for achieving full upper limb range of motion. To achieve normal upward rotation of the scapula, the serratus anterior must work together with the upper and lower fibers of the trapezius. Figure B shows how, when all three sets of fibers shorten, the result is upward scapular rotation.



Teres major:

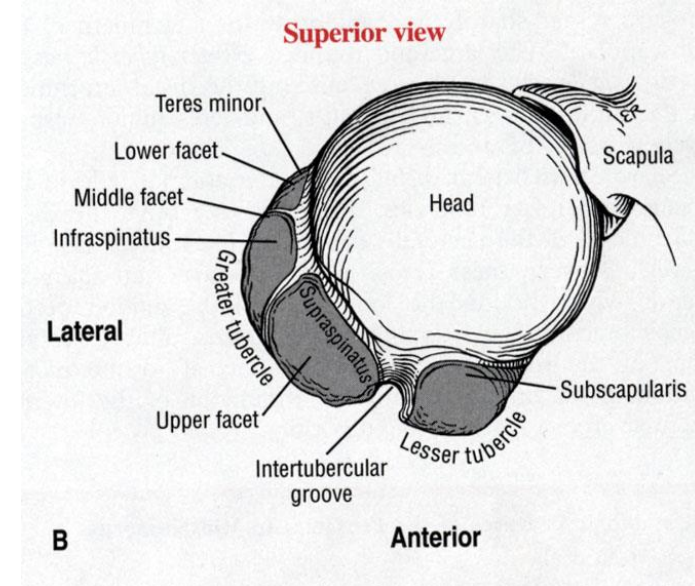
- **Origin:** from dorsal surface of inferior angle of scapula
- **Insertion:** intertubercular groove of humerus
- **Innervation:** lower subscapular nerve (C5-C6)
- **Action:** Think of teres major as a “baby” latissimus dorsi. The fibers run in the same direction from origin to insertion as the latissimus dorsi and they have the same insertion. Given this, hypothesize the actions of teres major? (Answer at bottom of slide);



Answer: Extends, internally rotates and adducts the humerus.

Intrinsic Shoulder Musculature (Dorsal Scapular muscles): Overview

- 4 Rotator cuff (RC) muscles (also known as **SITS** muscles)
- All RC muscles originate from a scapular fossa named for the muscle
- The tendons of all 4 RC muscles insert into the proximal humerus AND the shoulder joint capsule:
 - **Greater tubercle (GT)** of humerus
 - **Supraspinatus** muscle (upper facet):
 - **Infraspinatus** (middle facet):
 - **Teres minor** (lower facet):
 - **Lesser tubercle**
 - **Subscapularis**



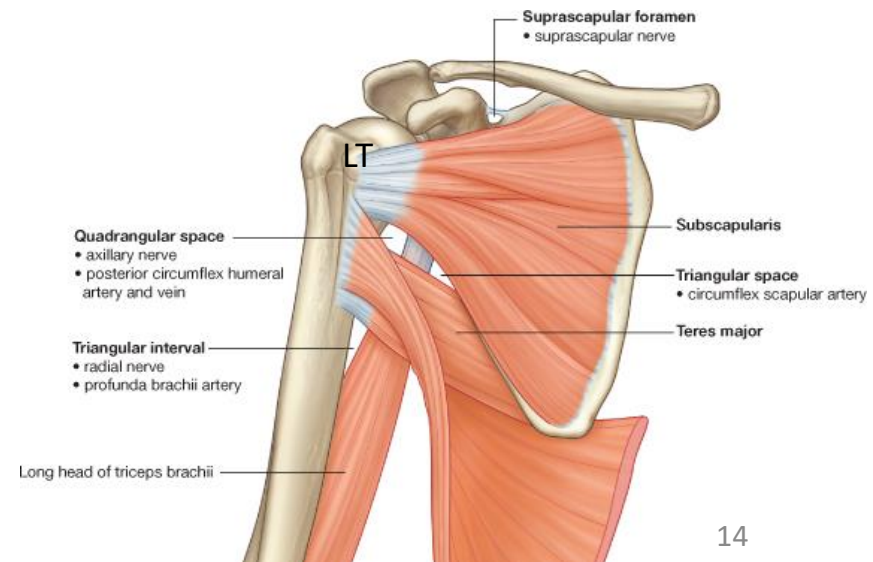
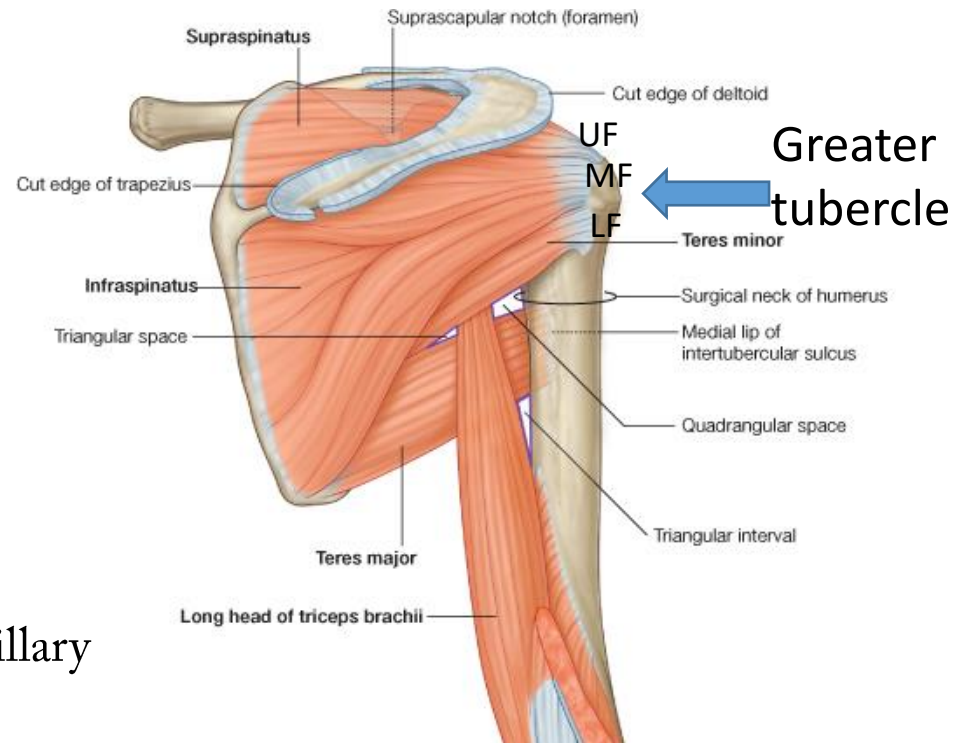
(Neumann DA, 2010)

Supraspinatus from **supraspinous fossa** to upper facet (UF) of the greater tubercle (GT) (**Innervation:** suprascapular (C5-C6))

Infraspinatus from **infraspinous fossa** to middle facet (MF) of GT (**Innervation:** suprascapular (C5-C6))

Teres minor from **teres minor fossa** to inferior facet (LF) of GT (**Innervation:** axillary nerve (C5-C6))

Subscapularis from subscapular fossa to **lesser tubercle (LT)** (**Innervation:** upper & lower subscapular nerves (C5-C6))

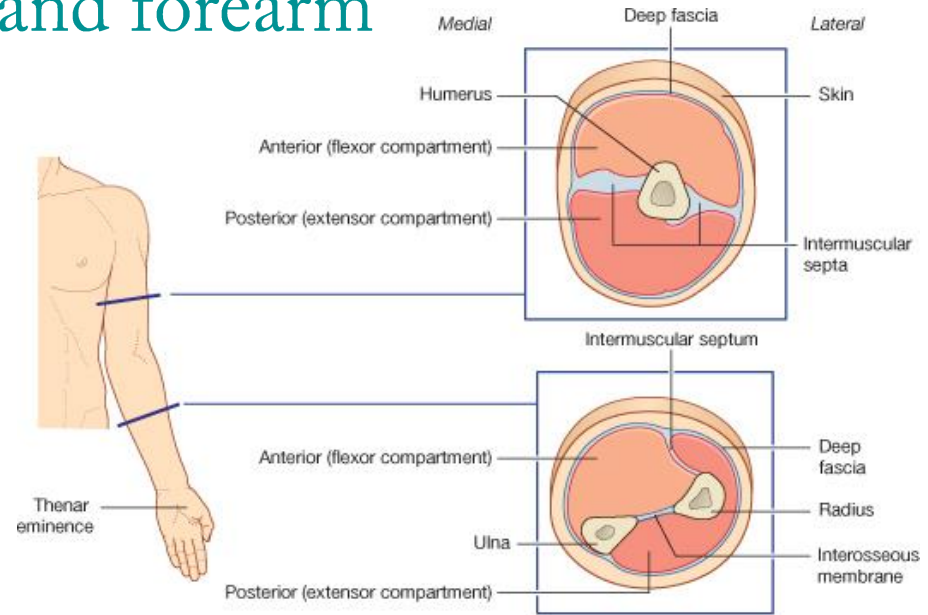


- Rotator cuff muscle actions:

- As a group, contraction of the rotator cuff muscles serves to “steer” the humeral head in the glenoid fossa, providing important positioning and stabilization of the humeral head in the glenoid fossa.
- Each muscle also has independent actions:
 - Supraspinatus muscle:
 - Primary function: Initiates first 15–20° of humeral abduction
 - Secondary function: Assists (slight) in external rotation of humerus
 - Infraspinatus muscle: External rotator of humerus
 - Teres minor muscle: External rotator of humerus
 - Subscapularis muscle: Primary internal rotator of humerus; also adducts the humerus.

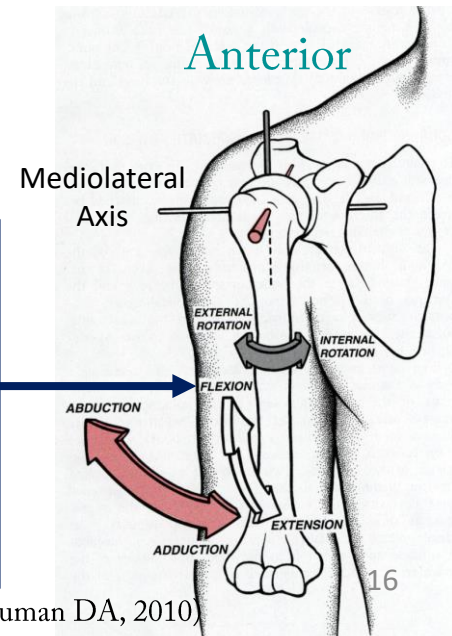
Organization of muscle compartments: Arm and forearm

- Arm and forearm muscles are separated into anterior and posterior compartments by fascia, bones and ligaments.
- The anterior compartments are referred to as “flexor” compartments because anterior compartment muscles are positioned anterior to the mediolateral axis for sagittal-plane motion. Thus anterior compartment muscles are “flexors”.
- Similarly, posterior compartment muscles are “extensors” because they lie posterior to the mediolateral axis for sagittal-plane motion.
- Realize that these muscles do much more than flex and extend the upper extremity!



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Anterior compartment muscles are positioned along the anterior surface of the arm and forearm. Thus, when muscles that cross the anterior elbow shorten, they flex the elbow (bring the forearm closer to the arm).



(Neuman DA, 2010)

Organization of muscle compartments: Arm

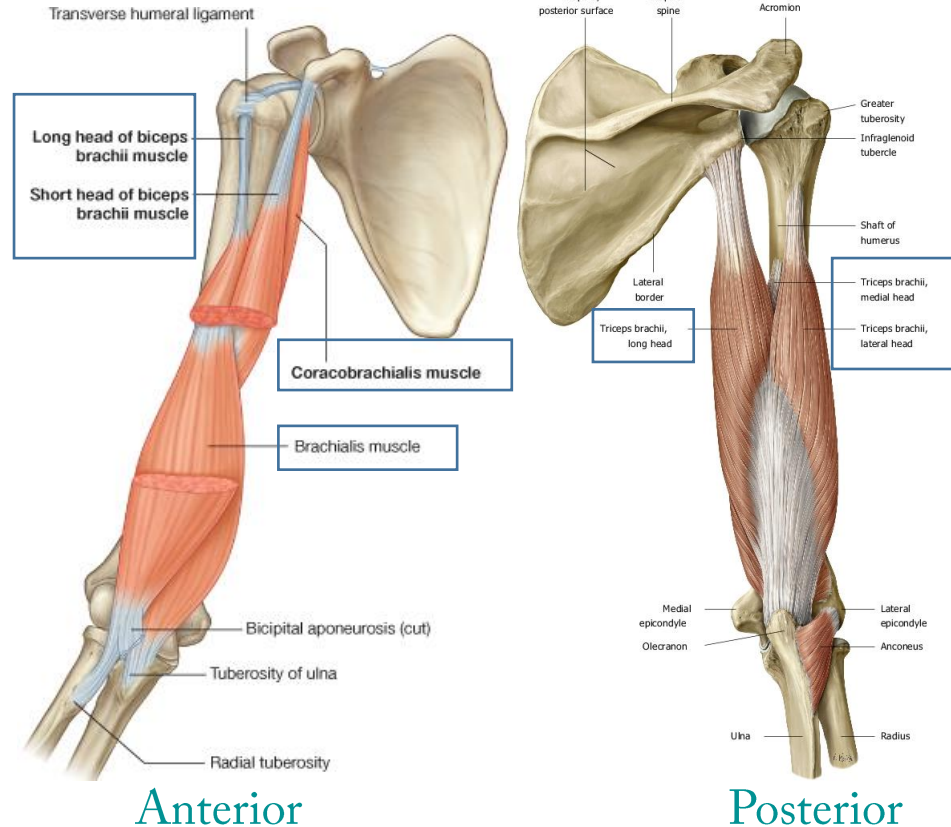
Organized into anterior and posterior compartments based on actions and innervation:

- **Anterior compartment muscles run anterior to humerus:**

- Act either on the humerus at the shoulder (coracobrachialis), the elbow (brachialis), or both (biceps brachii). Examples:
 - Shoulder flexion/internal rotation
 - Elbow flexion/supination
- All innervated by the same nerve: **musculocutaneous nerve (C5-C7)**.

- **Posterior compartment muscles run posterior to the humerus:**

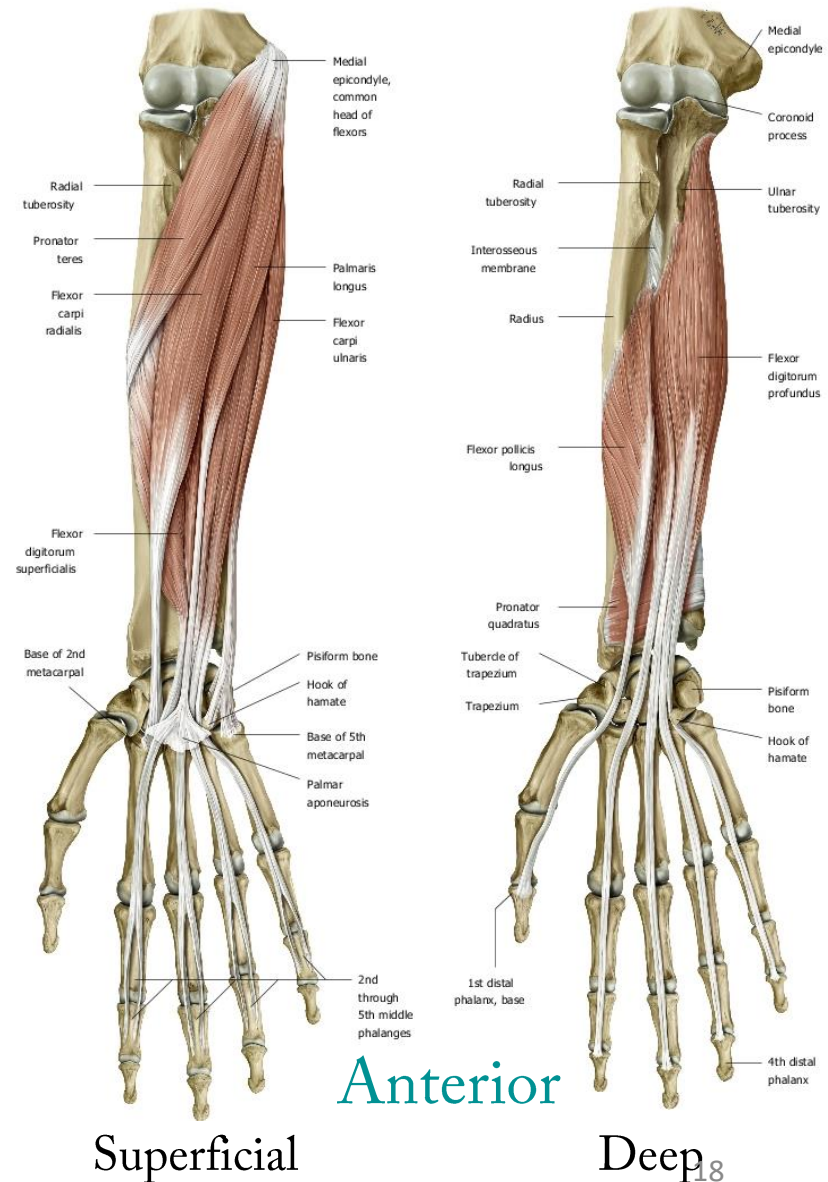
- Act either on the humerus at the shoulder, the elbow, or both. Examples:
 - Shoulder extension
 - Elbow extension
- All innervated by the same nerve: **radial nerve (C5-C8)**.



Organization of muscle compartments: Forearm

Organized into anterior and posterior compartments based on actions and innervation:

- **Anterior compartment muscles run anterior to the forearm:**
 - 8 muscles
 - Act on / at the elbow, wrist or hand.
Examples:
 - Elbow flexion
 - Wrist flexion/deviation
 - Digital flexion
 - All but 1 ½ muscles innervated by **median nerve (C6-T1)**; the other 1 ½ muscles (fl. carpi ulnaris & medial portion of fl. digit. prof.) innervated by **ulnar nerve (C8-T1)**.



Anterior Forearm Compartment: Flexors (8 muscles)

Superficial Layer (1)

- Pronator teres
- Flexor carpi radialis
- Palmaris longus
- Flexor carpi ulnaris

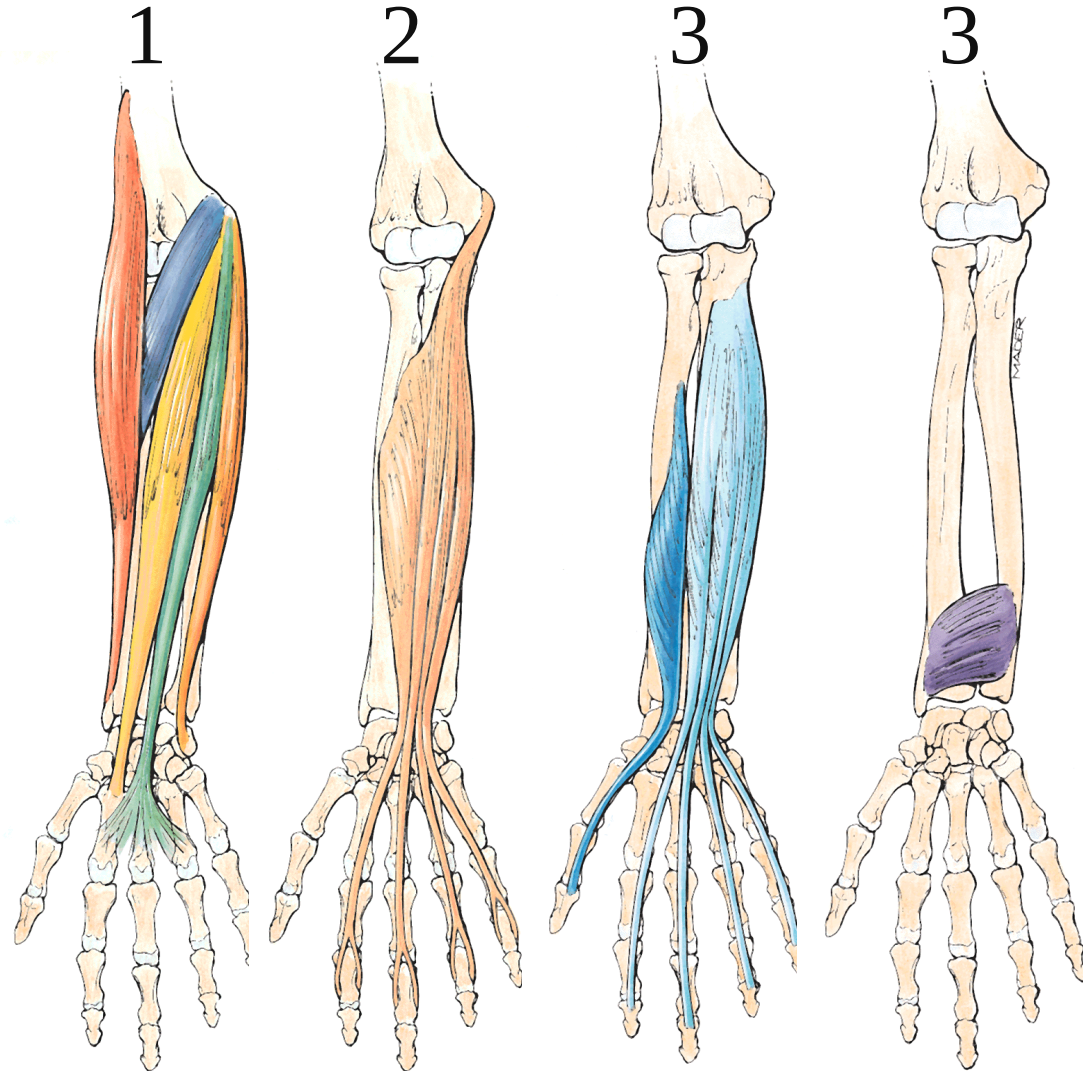
Intermediate Layer (2)

- Flexor digitorum superficialis

Deep Layer (3)

- Flexor digitorum profundus
- Flexor pollicis longus
- Pronator quadratus

Overall, these muscles provide for forearm pronation, and elbow, wrist, and finger (digital) flexion. Know which muscles perform which actions. The muscle-tendon units that run to the digits are known as **extrinsic hand muscles** because **they originate in the forearm and insert into the hand.**

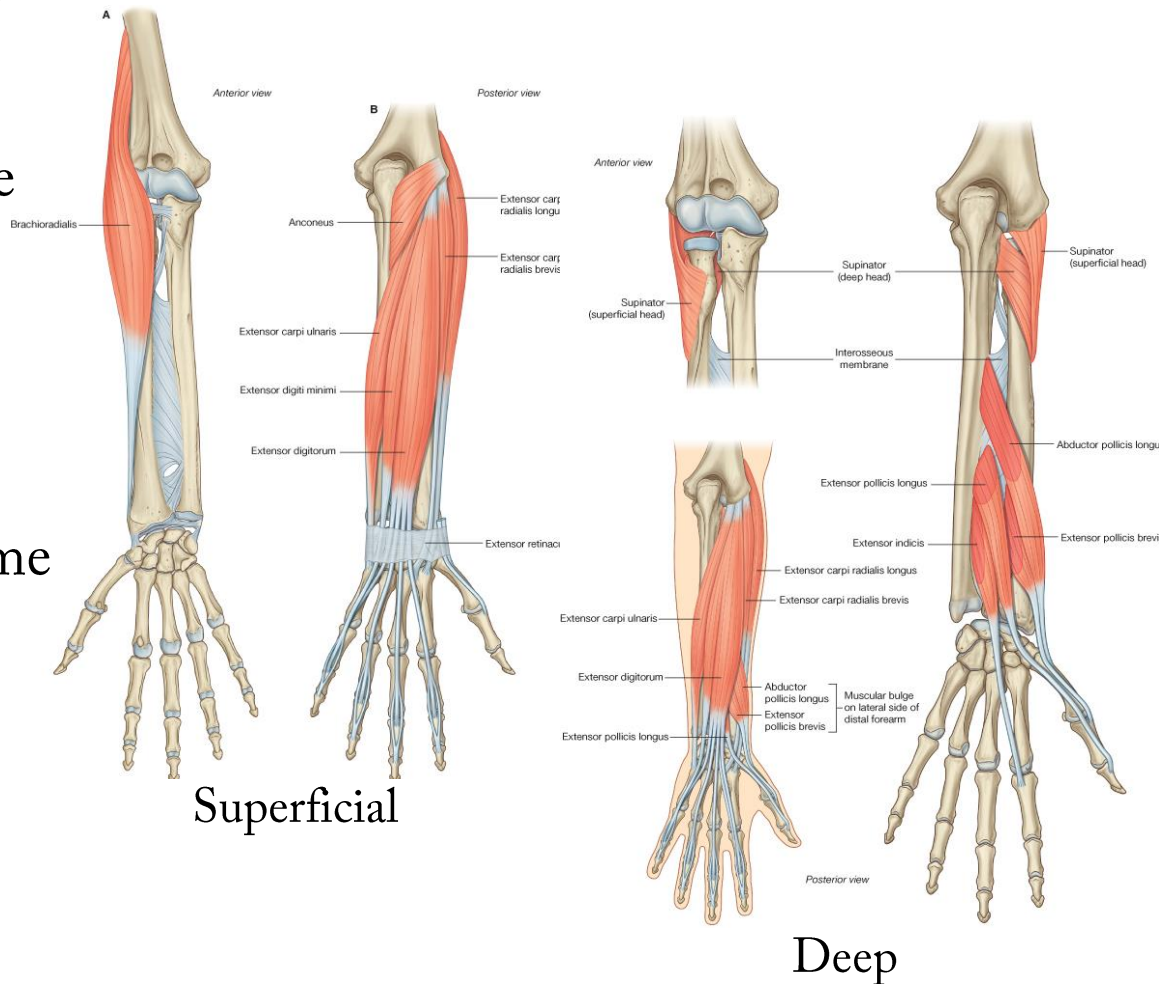


Anterior Views

Organization of muscle compartments: Forearm

- **Posterior compartment muscles run posterior to the forearm:**
 - 11 muscles
 - Act on / at the elbow, the wrist or the hand.
- Examples:
 - Elbow extension
 - Wrist extension/deviation
 - Digital extension
- All innervated by the same nerve: **radial n (C5-C8)**

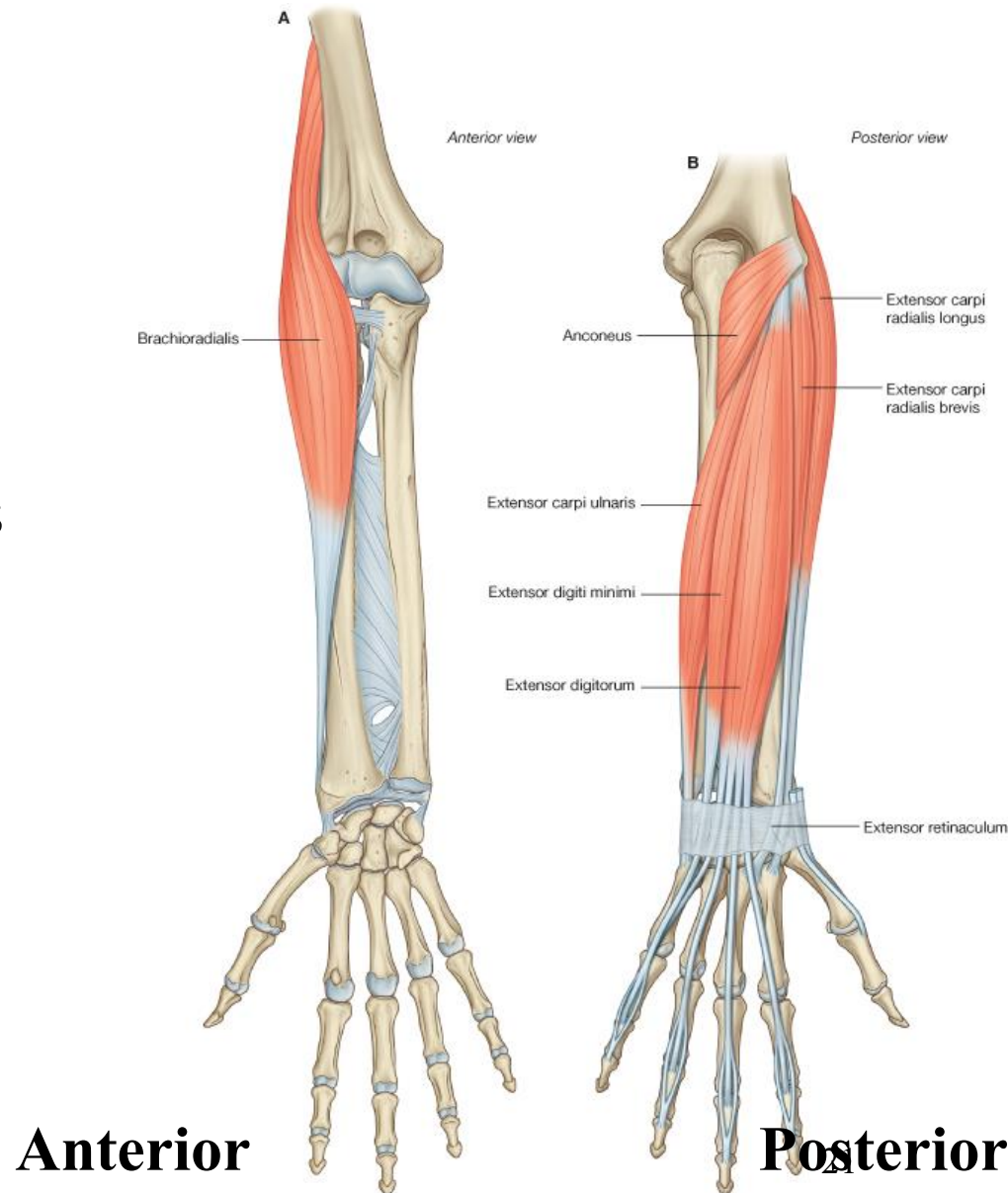
Posterior



Forearm Extensors

Superficial muscle group (6 muscles)

- brachioradialis
- extensor carpi radialis longus
- extensor carpi radialis brevis
- extensor digitorum communis
- extensor digiti minimi
- extensor carpi ulnaris



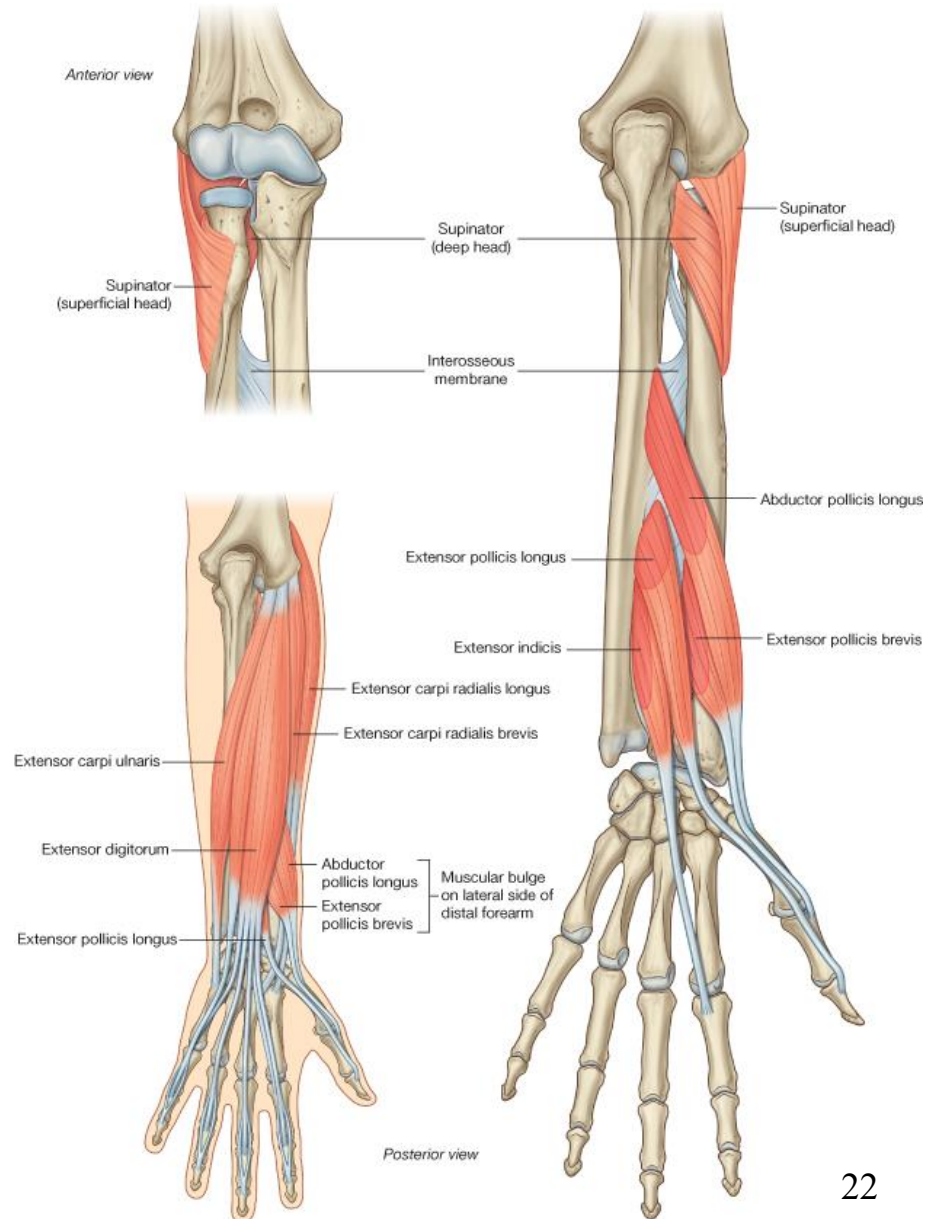
Forearm Extensors

Deep muscle Group (5 muscles)

- supinator
- abductor pollicis longus
- extensor pollicis brevis
- extensor pollicis longus
- extensor indicis

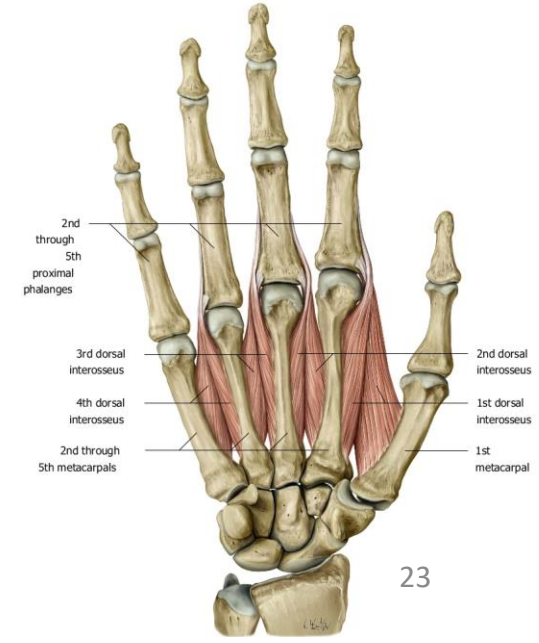
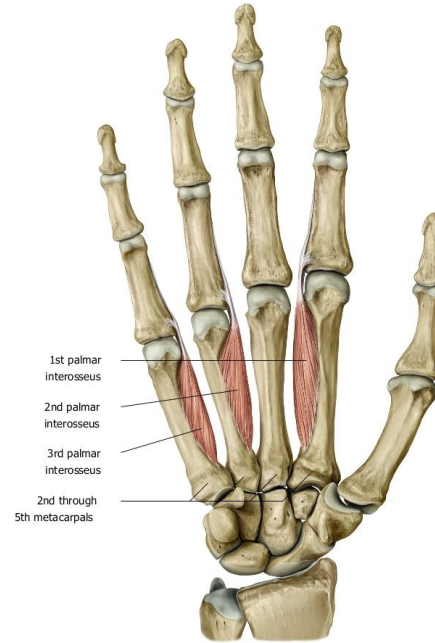
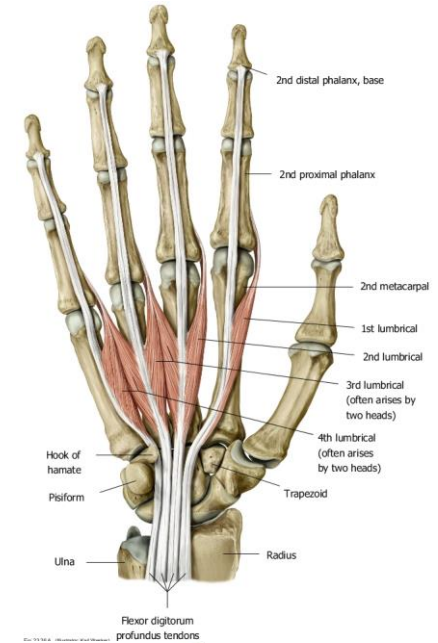
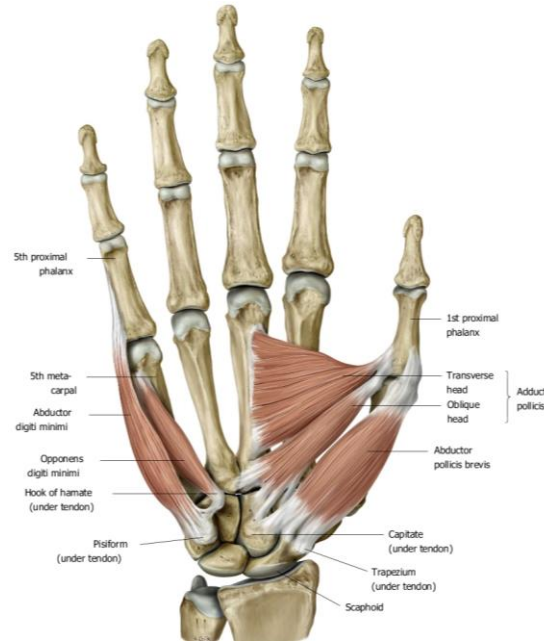
Overall, these 11 muscles provide for forearm supination, and elbow, wrist, and digital extension. Know which muscles do which.

Similar to the flexor muscles, the extensors that cross the wrist to insert into the hand are known as extrinsic hand muscles.



Organization of muscle compartments: Hand

- Intrinsic hand muscles = muscles that originate and insert into the hand on the ventral (palmar) surface (there are NO dorsal intrinsic hand muscles)
- These are small, short muscles that move the thumb and fingers.
- All innervated by **ulnar nerve (C8-T1)** except 4 muscles, which are innervated by **median nerve (C8-T1)** (L.O.A.F. – first two lumbricals, opponens pollicis, abductor pollicis brevis, flexor pollicis brevis).

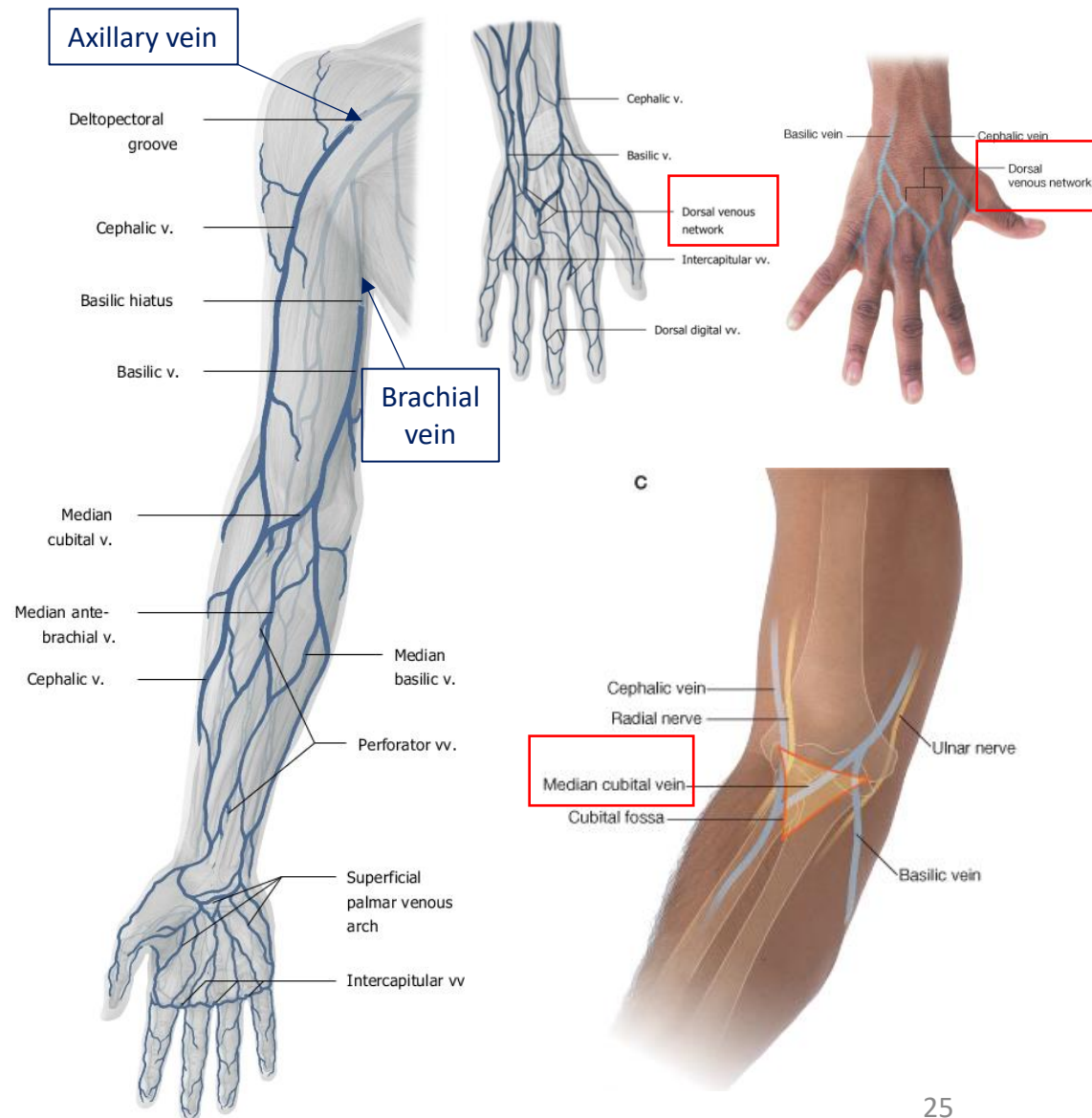


Muscle Compartments of Hand:

- **Thenar Compartment (C8-T1)** – Intrinsic muscles associated with movements of the thumb:
 - Abductor pollicis brevis
 - Flexor pollicis brevis
 - Opponens pollicis
- **Hypothenar Compartment (C8-T1)** – Intrinsic muscles associated with movements of the little finger
 - Abductor digiti minimi
 - Flexor digiti minimi brevis
 - Opponens digiti minimi
- **Central Compartment**
 - Long digital flexor tendons (FDS/FDP and FPL and their synovial tendon sheaths)
 - Lumbricals (C8-T1)
- **Adductor/Interossei Compartment**
 - Adductor pollicis + dorsal and palmar interossei (C8-T1)

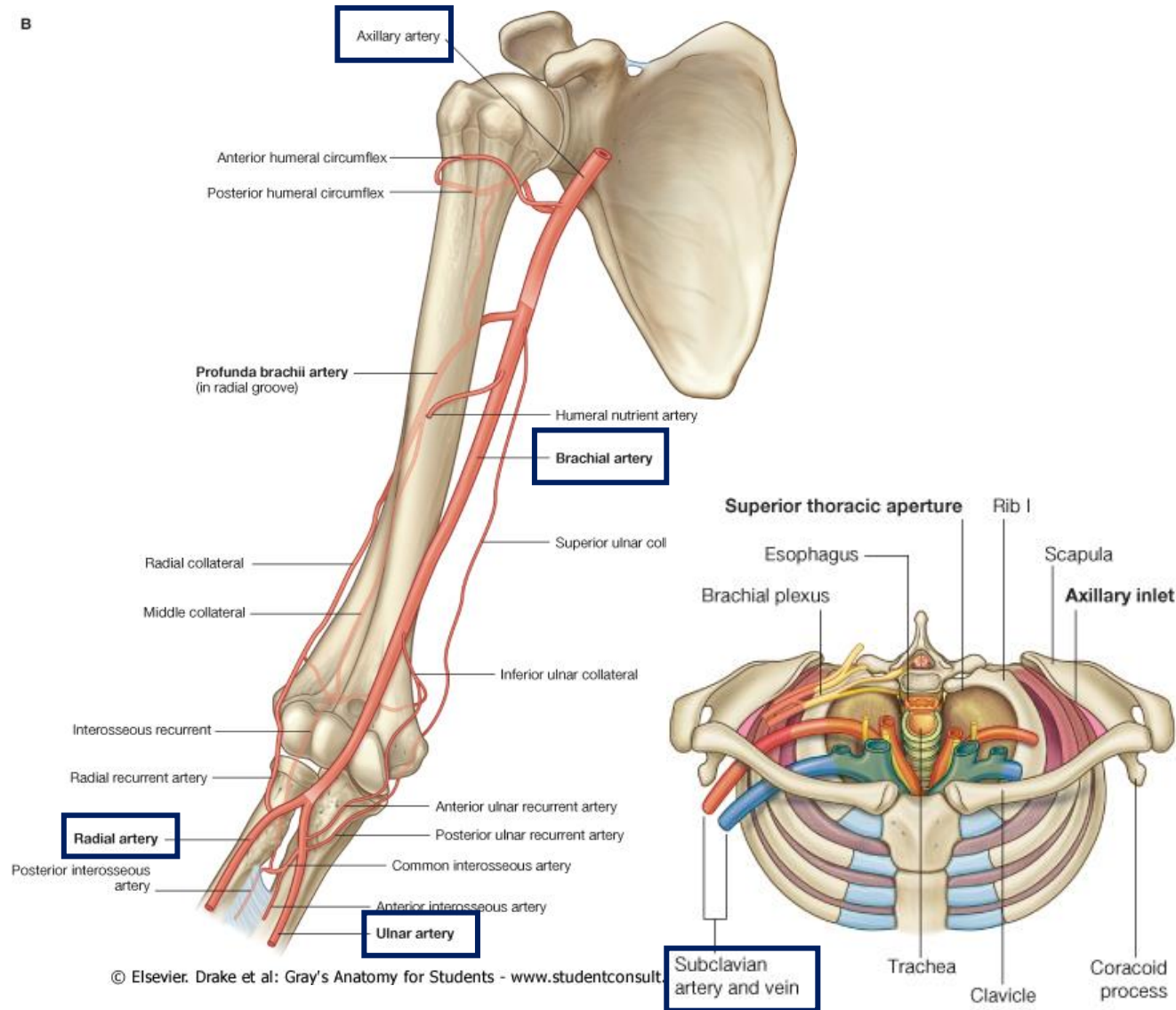
Upper Extremity: Superficial Veins

- Large veins embedded in the superficial fascia that drain blood from distal to proximal and from superficial to deep.
- Chief superficial UE veins are the cephalic (lateral) and basilic (medial) veins, which originate from the dorsal venous network of the hand.
- The cephalic v. drains directly into the **axillary v.**, a deep vein of the UE. The basilic v. generally drains directly into the **brachial v.**, a deep vein that becomes the axillary v.
- The cephalic and basilic vv. form a connection at the cubital fossa (anterior elbow) called the median cubital vein. This vein is frequently used to draw blood or establish an IV.



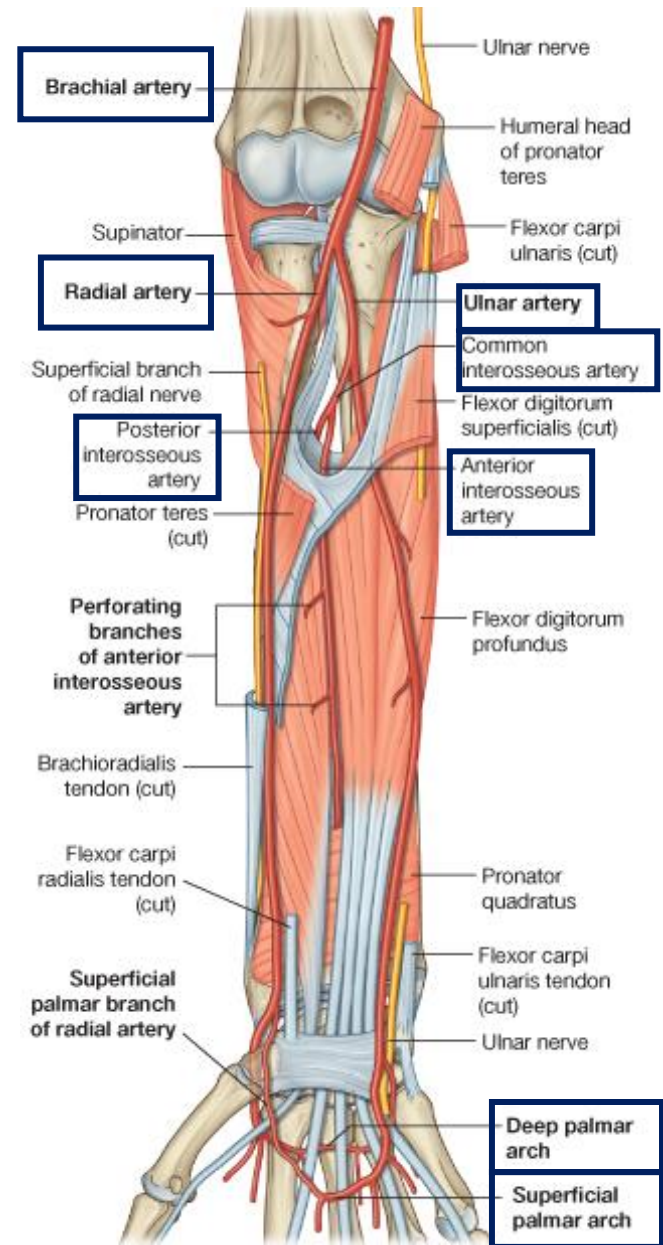
Upper Extremity: Vasculature

- The major blood supply to the upper extremity derives from one long artery that changes its name as it passes a series of landmarks. Realize that this artery runs with paired (often same-named) veins.
- Blood to the UE begins with the **subclavian artery** which changes its name to axillary artery when it reaches the lateral border of the 1st rib.
- Axillary artery** serves the region of the axilla. It then becomes the brachial artery when it enters the arm (at the lower border of the teres major muscle- see Gray's p. 733).
- Brachial artery** serves the region of the arm. When it crosses the cubital fossa, it generally splits into two main branches: radial artery and ulnar artery.
- Radial and ulnar arteries** serve the forearm.



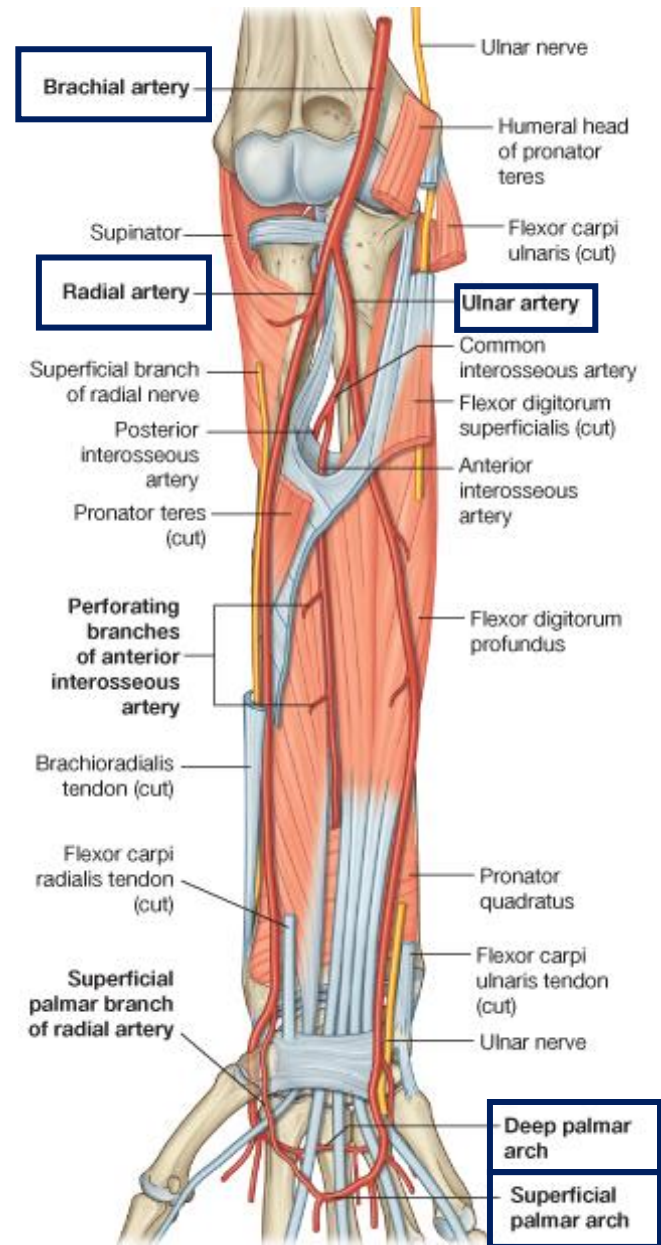
Upper Extremity: Vasculature

The radial and ulnar arteries generally appear at the elbow (although they can appear much more proximally in the arm). The radial a. stays in the anterior compartment of the forearm, as does the main trunk of the ulnar a. However, the ulnar a. also gives rise to the **common interosseous a.**, which then gives rise to the anterior and posterior interosseous aa. The **anterior interosseous a.** stays in the anterior compartment while the **posterior interosseous a.** passes superior to the interosseous membrane and provide the bulk of the blood supply to the posterior compartment of the forearm.

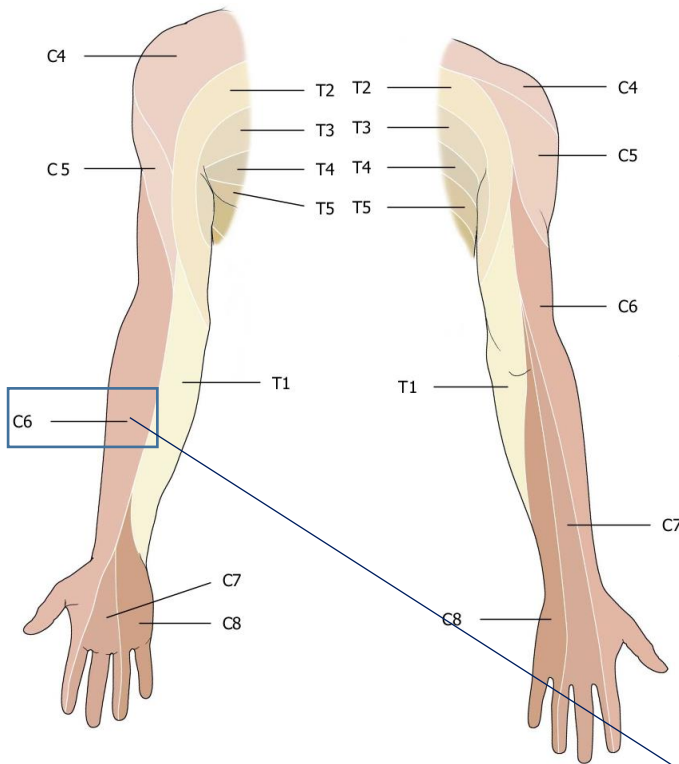


Upper Extremity: Vasculature

The radial and ulnar arteries cross the wrist and contribute to the formation of the **superficial and deep palmar arterial arches**, which are the main source of blood to the hand.

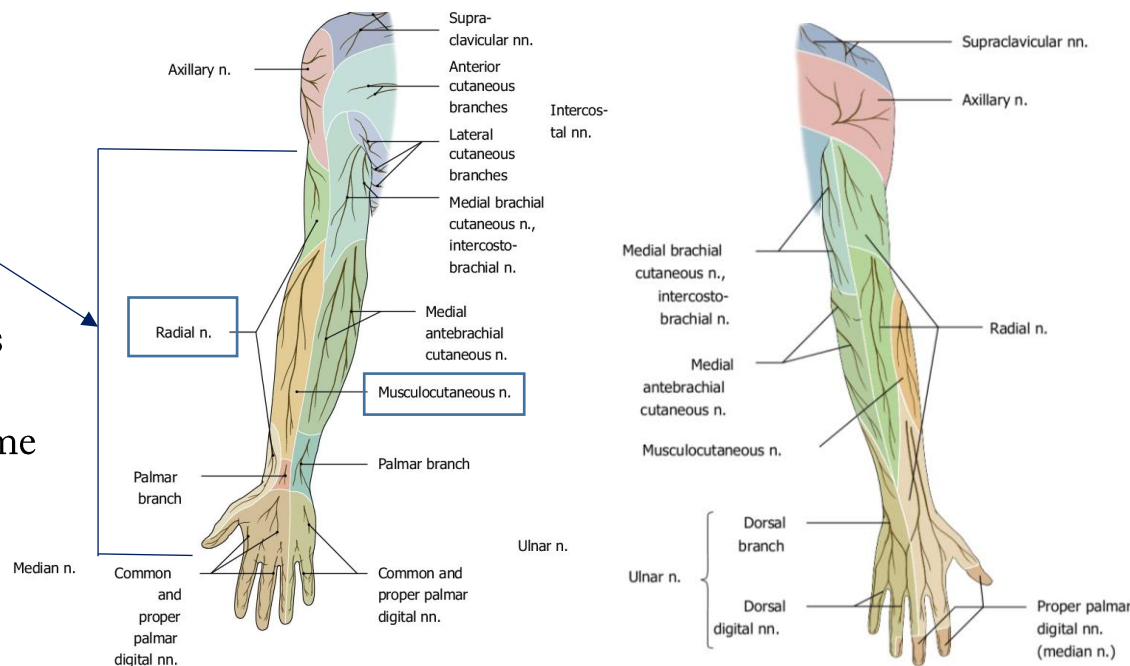


Upper Extremity: Dermatomes and cutaneous innervation



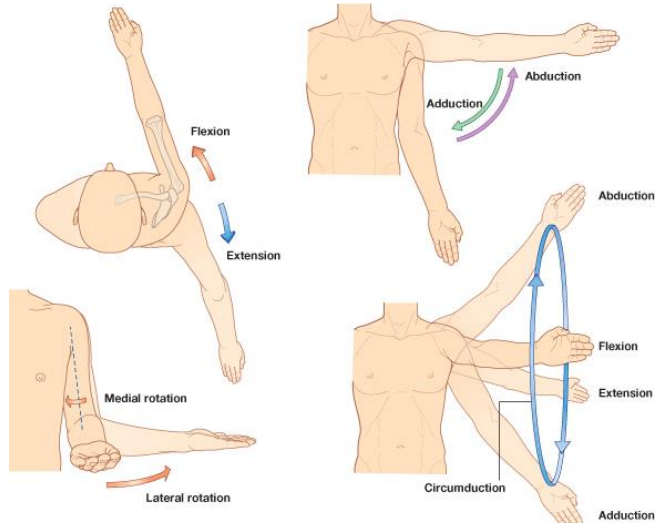
Dermatomes of the upper limb. Remember, a dermatome is the sensory innervation of a strip of skin from a single spinal nerve. So a strip of skin labeled “C6” is that portion of skin served by the sensory portion of the 6th cervical spinal nerve. Dermatomes are “maps” of the distribution of spinal nerves to the skin; dermatomes cannot be “dissected”.

A **variety** of nerve branches provide cutaneous innervation to the skin of the upper limb. Notice that multiple cutaneous nerves can serve the region of a single dermatome. For example, the C6 dermatome is served by branches of both radial and musculocutaneous nerves.

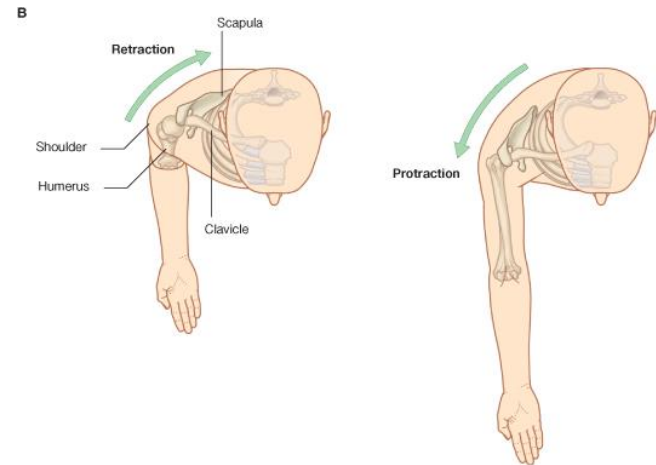


Upper Extremity Motions

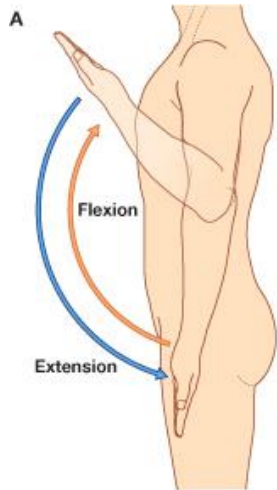
Glenohumeral Joint



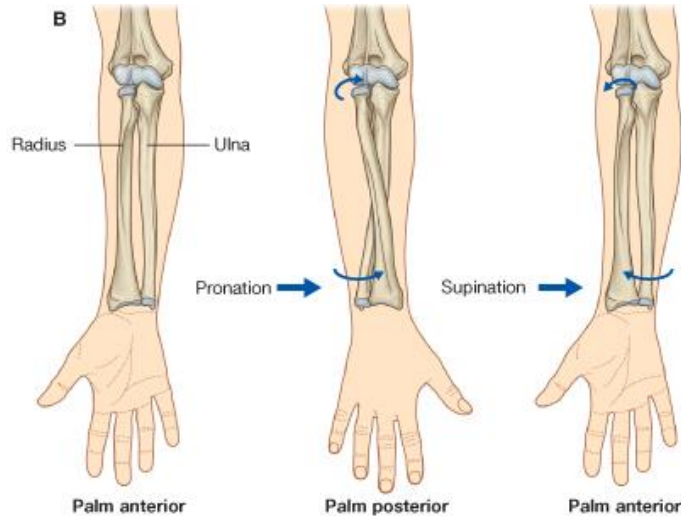
"Shoulder"



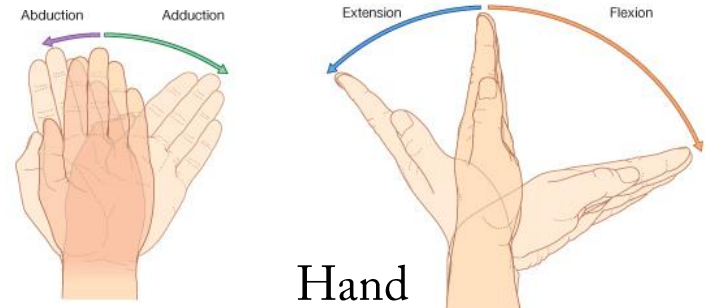
Elbow



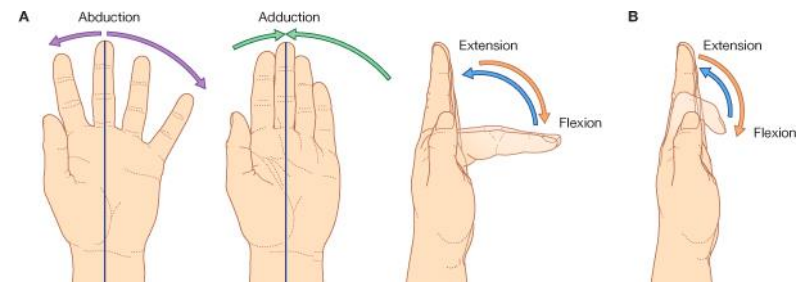
Forearm



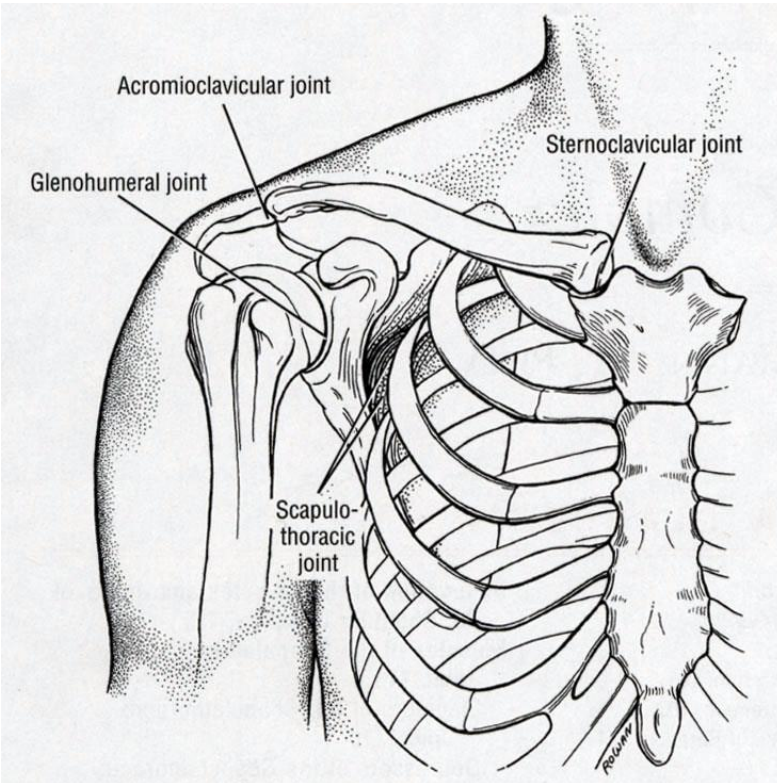
Wrist



Hand



Shoulder Joint Complex



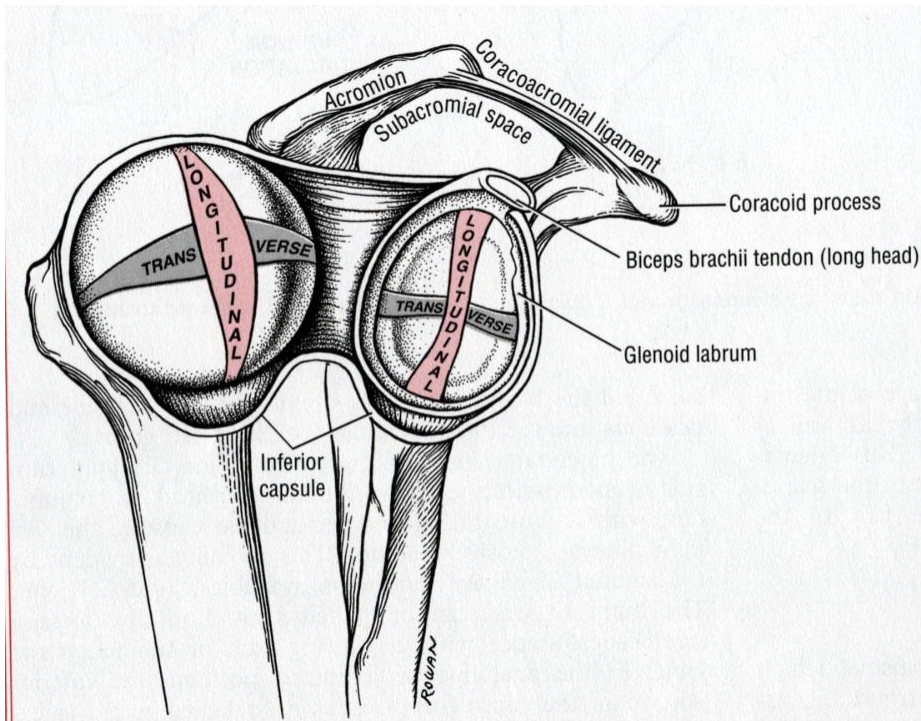
- Shoulder complex comprises humerus, scapula, clavicle, sternum & 1st rib.
- There are 3 or 4 joints to the shoulder complex (depending on how you want to think about them):
 - Three are anatomic (physiologic) joints:
 - Glenohumeral joint
 - Sternoclavicular joint
 - Acromioclavicular joint
 - One is a “functional” joint:
 - Scapulothoracic joint
- The shoulder complex is designed to facilitate upper extremity mobility.

GLENOHUMERAL JOINT

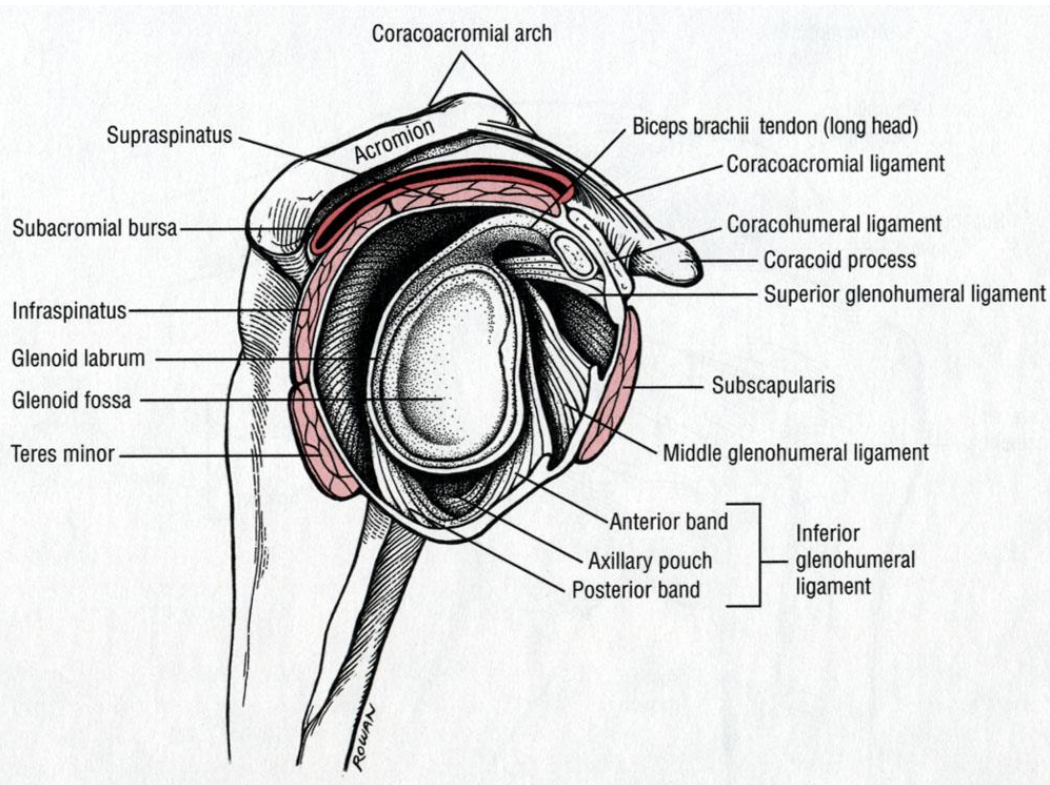


- The glenohumeral joint is a ball & socket joint.
 - Flexion/extension
 - Abduction/adduction
 - Internal(medial) /external (lateral) rotation
 - Circumduction is a composite of osteokinematic motions.
- Supported by:
 - Joint capsule
 - Ligaments

- Articular surfaces of the glenohumeral joint:

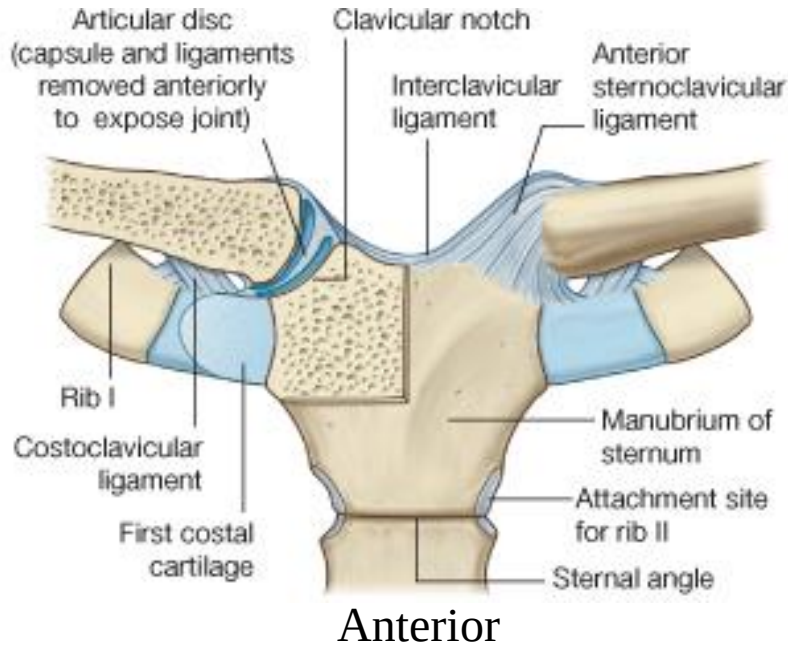


- **Glenoid fossa:** long longitudinally, narrow transversely, and shallow, covering only about 1/3rd of the articular surface of the humeral head.
- **Humeral head:** large in size and spherical in shape; the longitudinal diameter is ~2x greater than the corresponding surface of the glenoid fossa so that only a small portion contacts the glenoid fossa at any time.



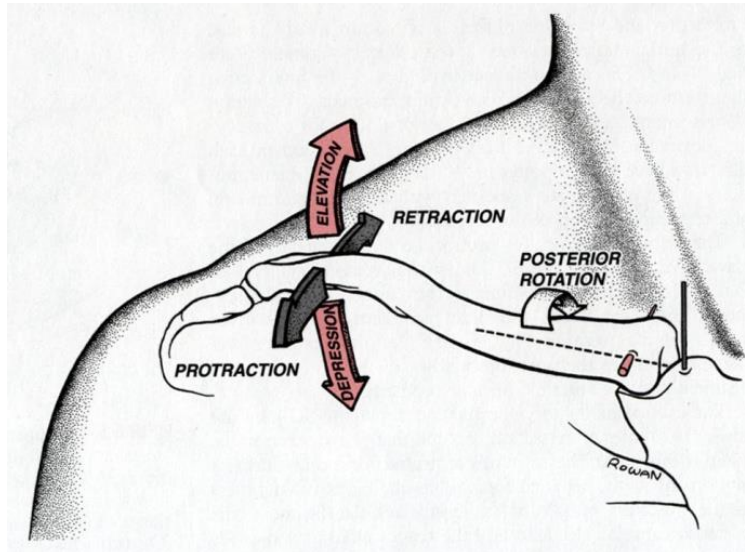
- GH joint is a classic synovial joint
- Associated structures (superficial to deep):
 - **Joint capsule***
 - Outer fibrous capsule
 - Inner synovial membrane
 - Tendon of the long head of the biceps brachii muscle
 - Glenohumeral ligaments
 - Glenoid labrum
 - Glenoid fossa

Sternoclavicular Joint



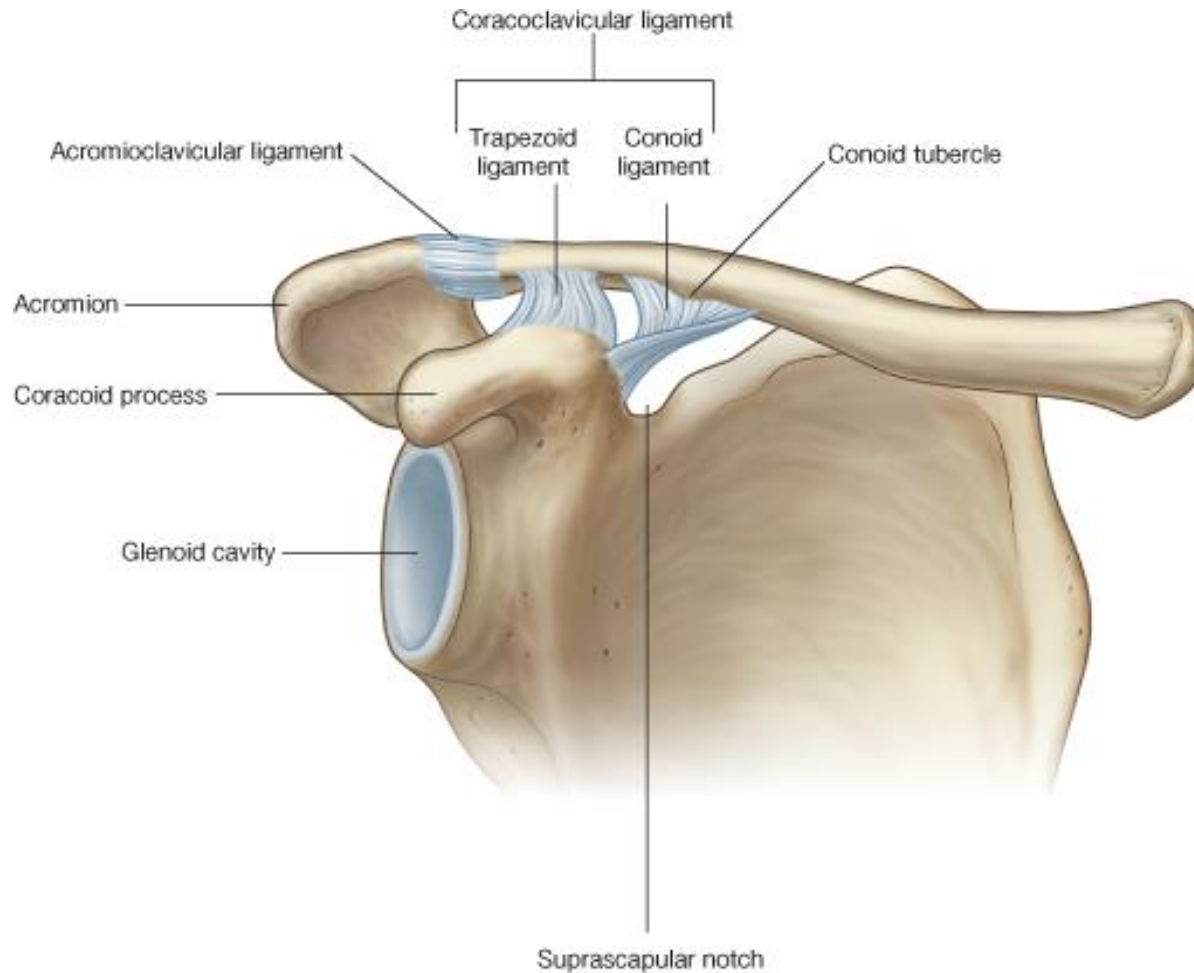
The sternoclavicular (SC) joint provides the only direct attachment of the upper limb to the axial skeleton.

A **saddle-shaped** synovial joint between the proximal clavicle and the **clavicular notch** of the **manubrium** of the sternum and 1st costal cartilage. A fibrous articular disc is interposed between the joint surfaces.

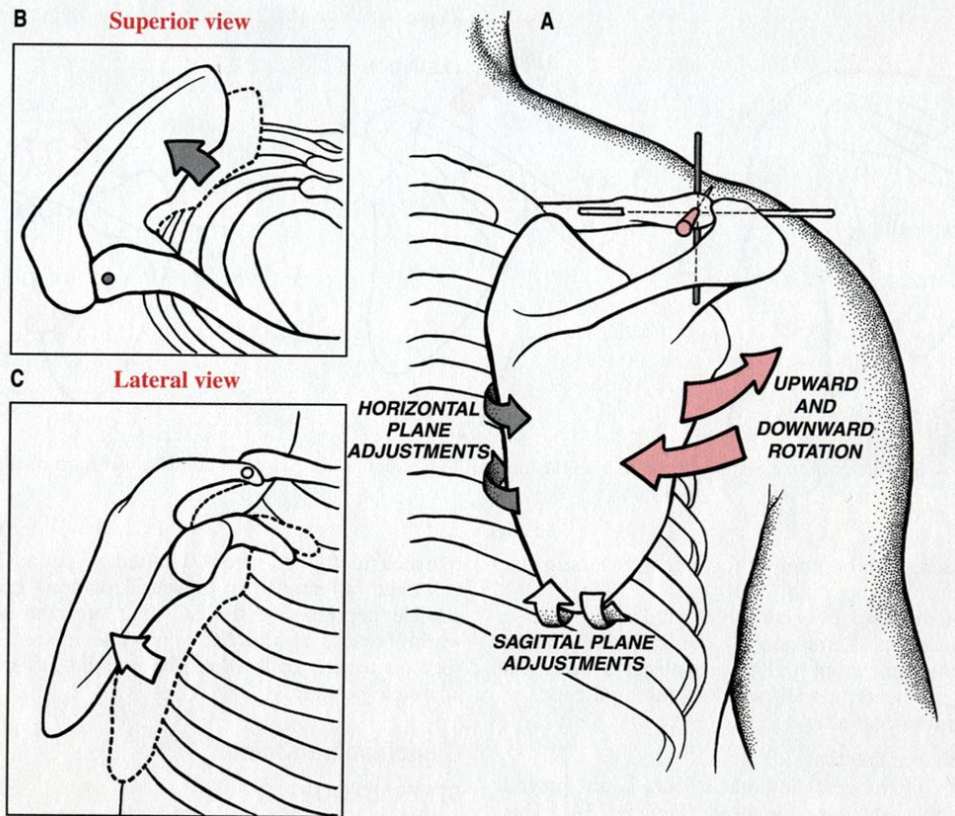


- **Elevation/depression**
 - Elevation: Distal clavicle elevates (proximal clavicle depresses).
 - Depression: Distal clavicle depresses (proximal clavicle elevates).
- **Protraction/retraction**
 - Protraction: Both ends of the clavicle move anteriorly.
 - Retraction: Both ends of the clavicle move posteriorly.
- **Axial rotation**
 - Clavicle rotates about its long axis.

Acromioclavicular Joint



The acromioclavicular joint is a plane synovial joint between the acromion process of the scapula and the distal (lateral) end of the clavicle. A fibrocartilaginous disc is interposed between the two articular surfaces.



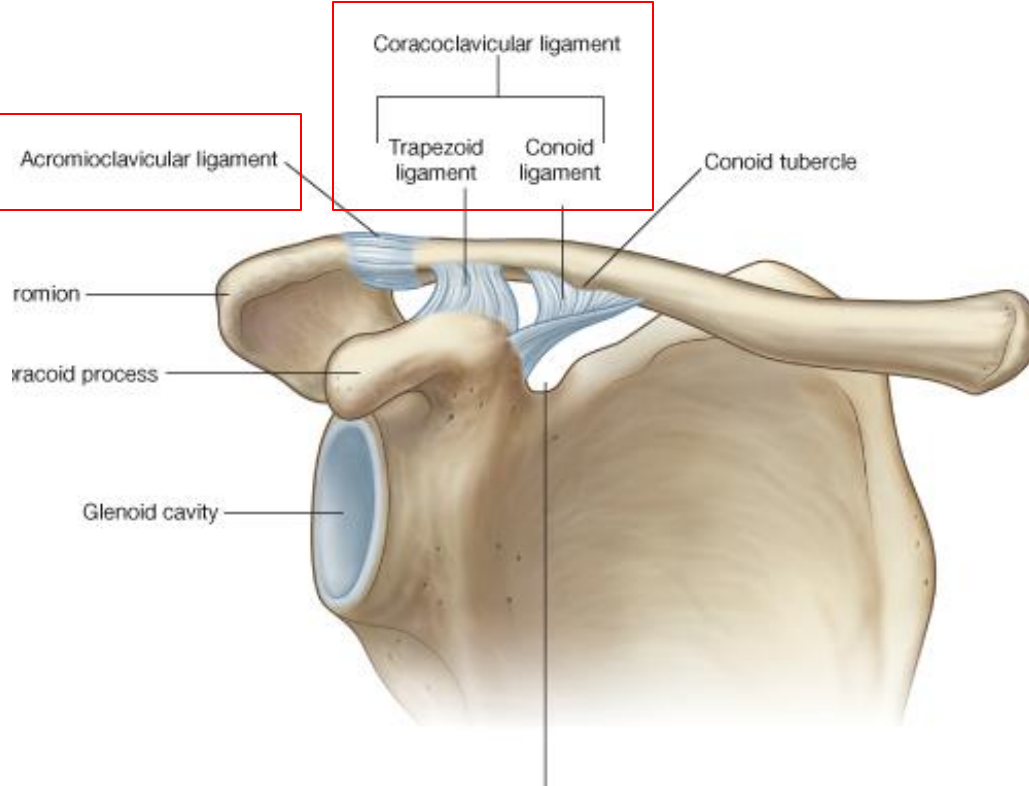
The AC joint provides for very small rotational movements of the acromion on the clavicle during scapular motion on the thorax.

These small rotational motions at the AC joint enable the scapula to move on the thorax:

1. elevate/depress (sagittal plane/mediolateral axis at AC)
2. rotate upward/downward (coronal plane/AP axis at AC)
3. protract/retract (horizontal plane/longitudinal axis at AC)

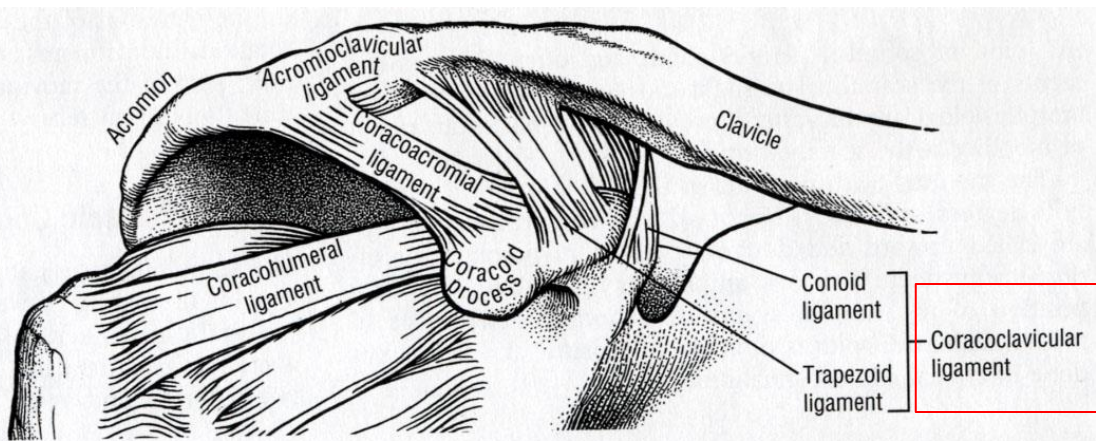
Ligamentous support

Superior and inferior AC ligaments reinforce the AC joint and help control movement of the acromion on the clavicle during motion at the AC joint.



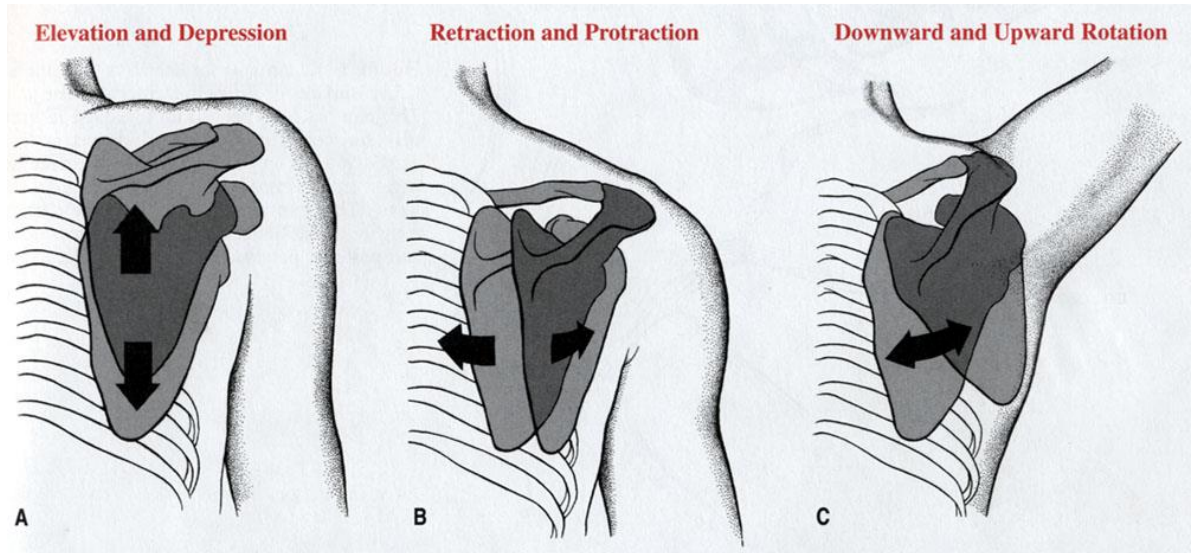
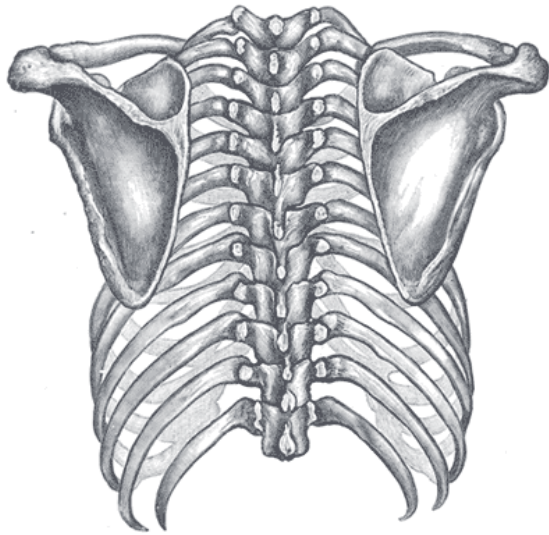
Coracoclavicular ligaments

- **Trapezoid fibers:** broad, quadrilateral-shaped bundle running anteromedially from clavicle to coracoid process of scapula.
- **Conoid fibers:** dense, triangular, posteromedial band from clavicle to coracoid process of scapula.
- Important for controlling **upward scapular rotation** during humeral elevation.

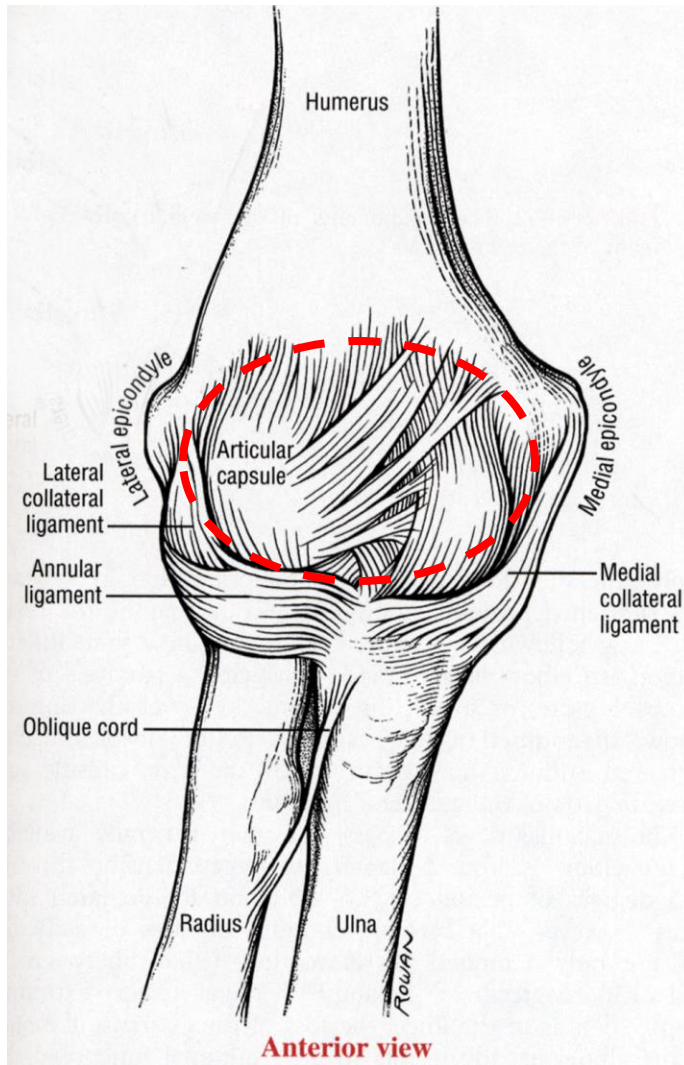


Scapulothoracic Joint

(i.e., movement of scapula relative to the thoracic wall)



Elbow Joint



The elbow joint is a **compound synovial joint** (compound meaning, more than two bones contribute to the joint).

Comprises the articular surfaces of the **distal humerus, proximal ulna, and proximal radius**, all surrounded by a single fibrous joint capsule.

The elbow complex is stabilized and reinforced by:

- bony articulation between the **olecranon process** and **olecranon fossa** of the ulna
- fibrous joint capsule
- **collateral ligaments**: medial (ulnar) and lateral (radial) collateral ligaments.

Joints of the **elbow** include the:

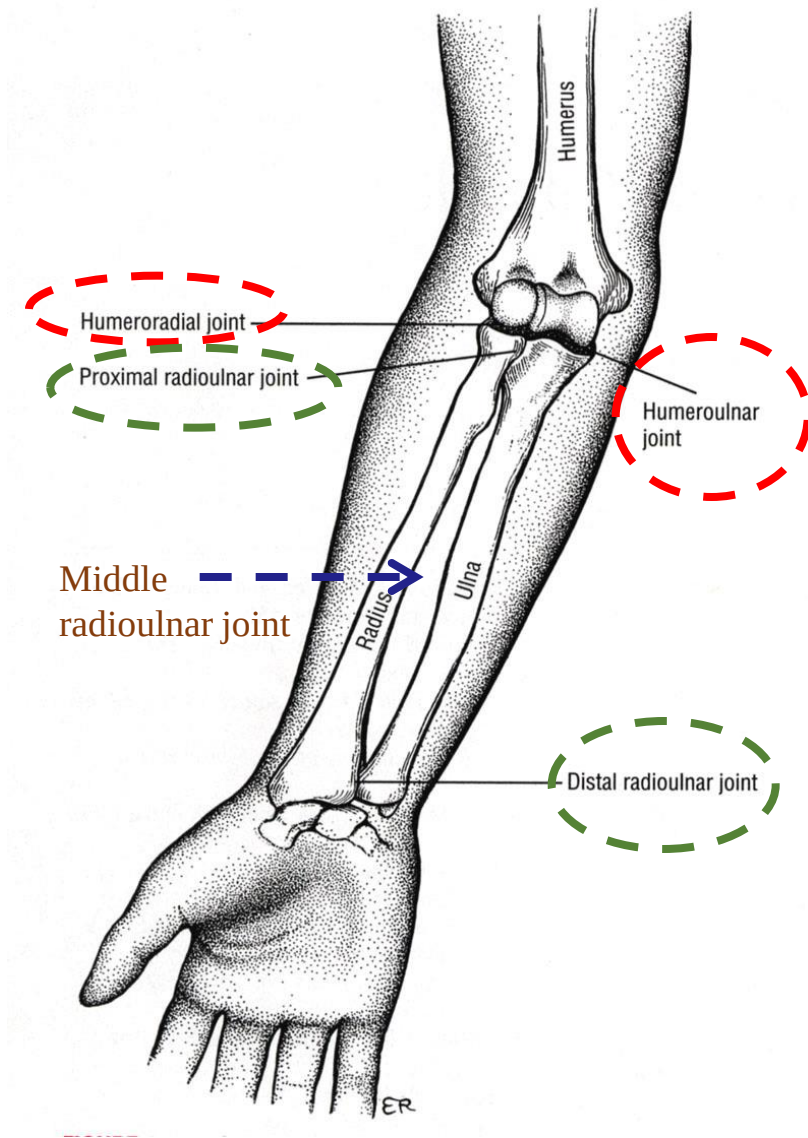
- **Humeroulnar joint**
- **humeroradial joint**

Joints of the **forearm** include the:

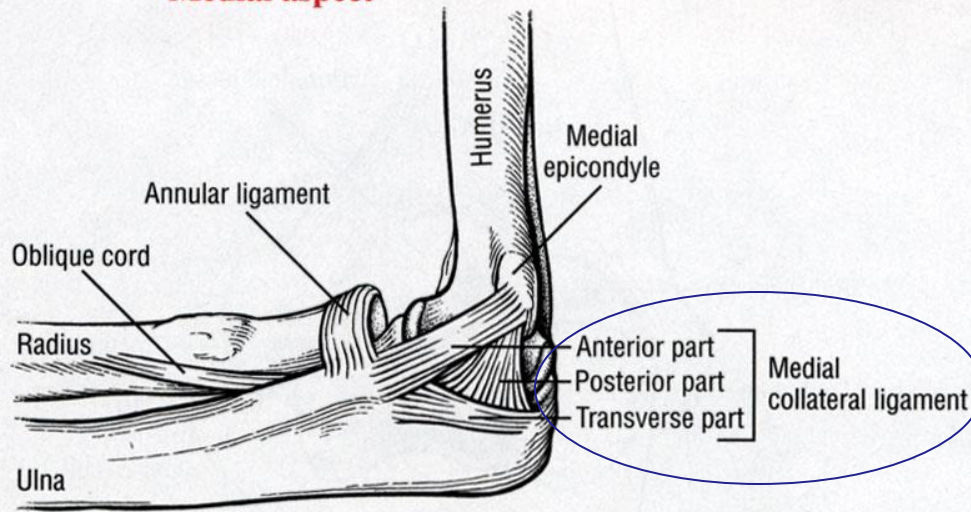
- **Proximal radioulnar joint (synovial)**
- **Middle radioulnar joint** (a fibrous joint between the crests of the radius and ulna)
- **Distal radioulnar joint (synovial).**

Together, the elbow and forearm joints function to:

- adjust and increase the functional range of the upper limb
- allow for pronation/supination of the forearm independent of shoulder movement, for greater functional flexibility and manipulation.



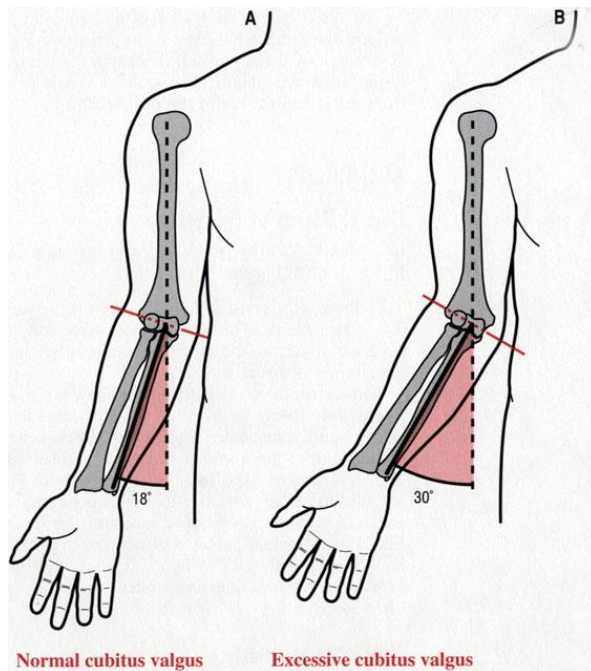
Medial aspect



The humeroulnar and humeroradial joints are reinforced medially and laterally by collateral ligaments.

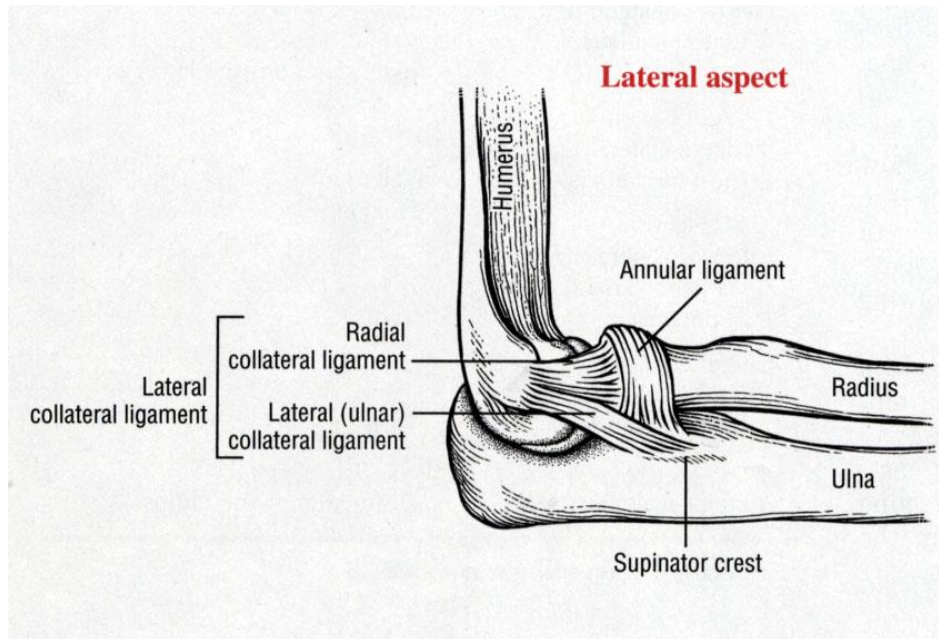
Medial collateral ligament (MCL):

- Runs from the medial epicondyle of the humerus to the coronoid process and olecranon process of the ulna.
- Collectively the medial collateral ligament provides the primary limit to a valgus force i.e., a force that would increase the cubital angle.



Given enough time, laxity of the MCL can lead to:

- Repetitive tension on the MCL
- Compression at the lateral joint (degenerative joint disease) between the radial head and capitulum of humerus
- Instability at medial joint
- Excessive cubitus valgus

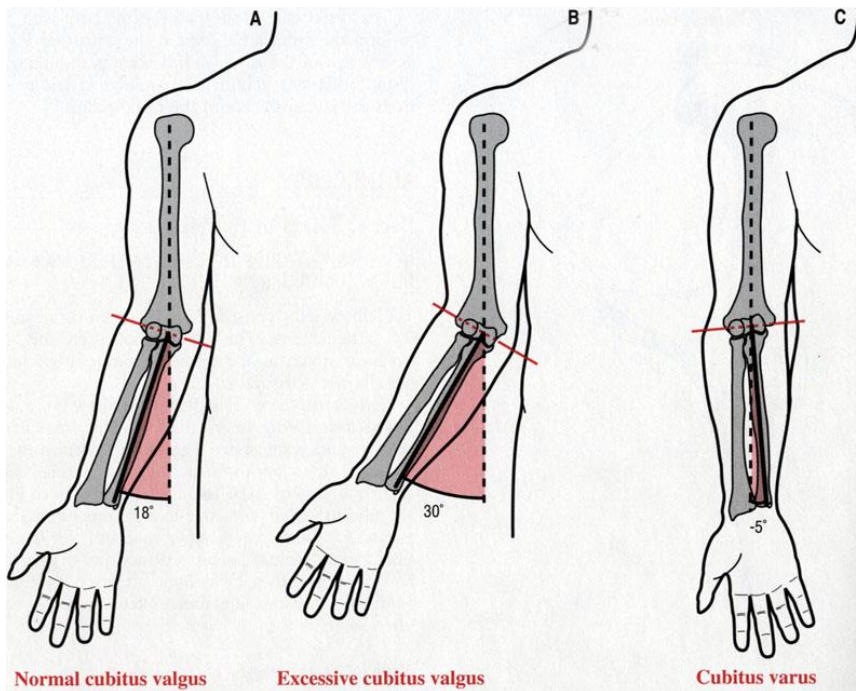


Lateral collateral ligament

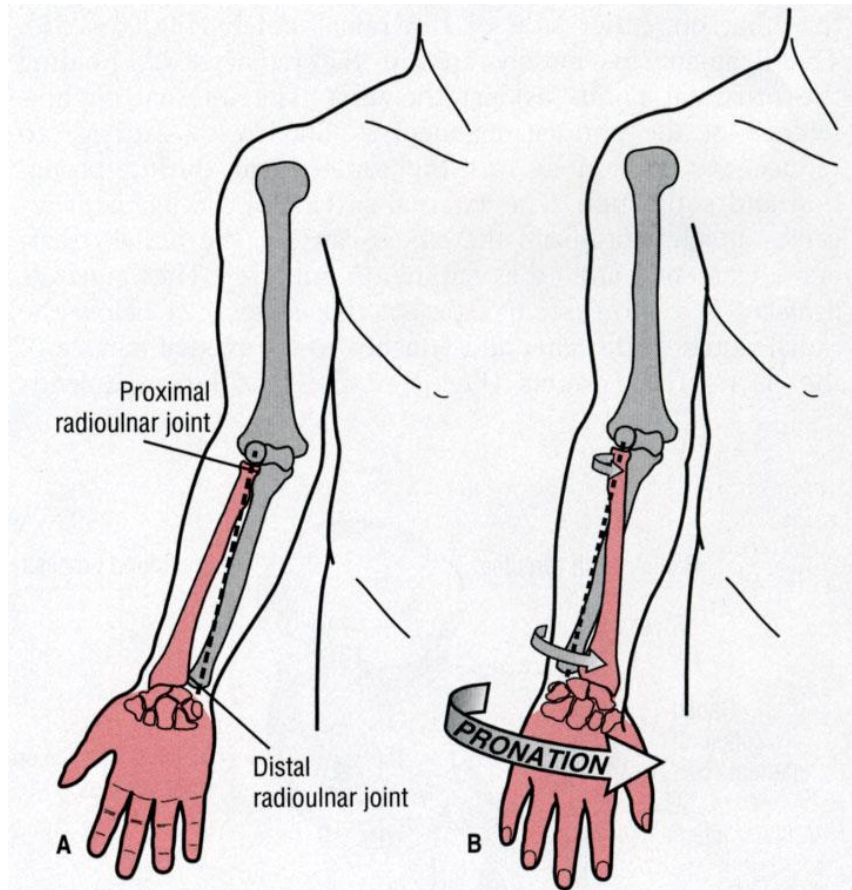
The lateral collateral ligament (LCL) is weaker than the medial collateral ligament.

Runs from the lateral epicondyle of the humerus to the annular ligament of the radius and to the proximal ulna.

Provides the primary limit to a varus directed force (a force that would lead to a decrease in the cubital angle).



Joints of the Forearm



Supination

(Neumann, 2002)

The joints of the forearm are all designed to:

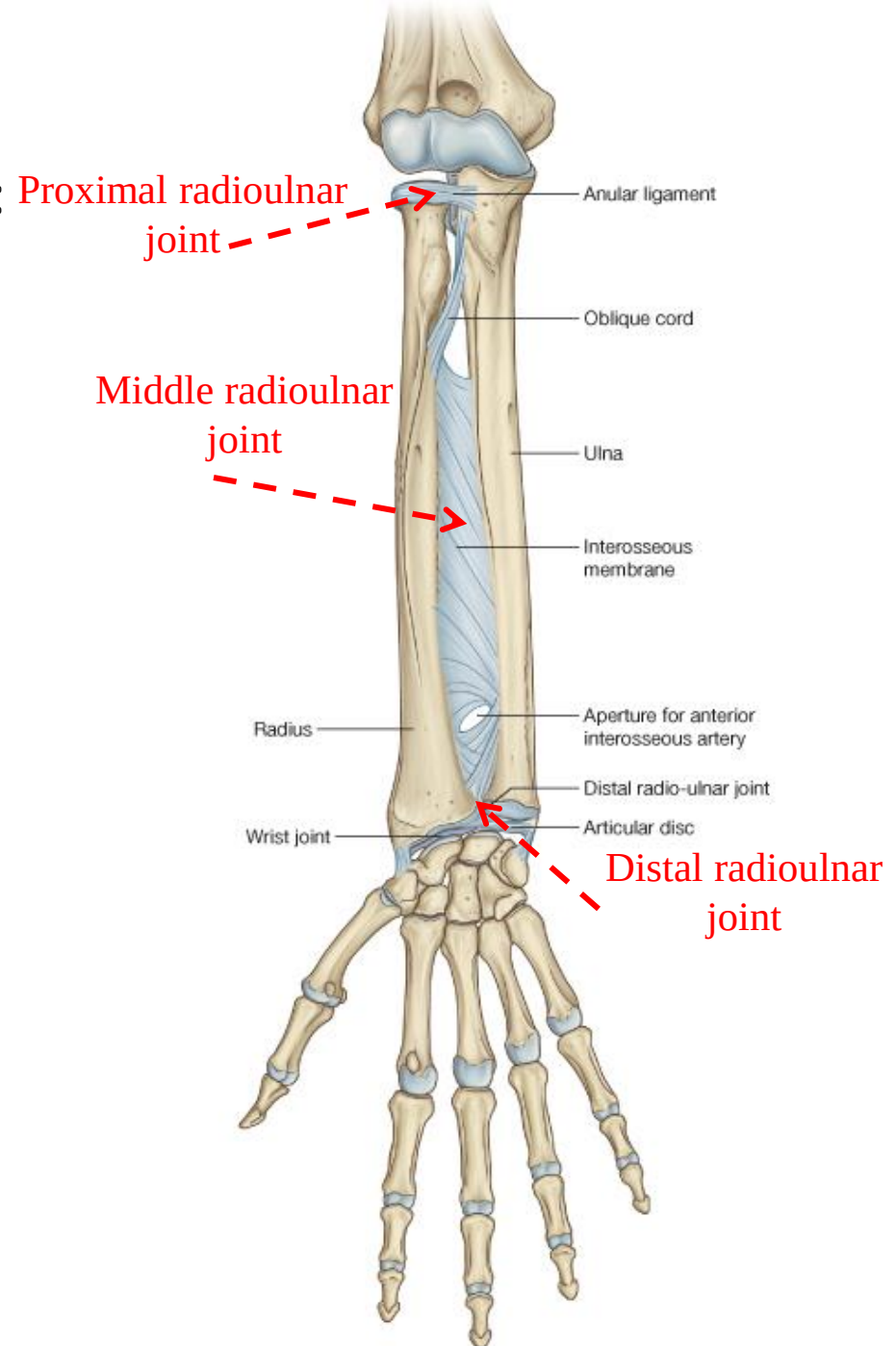
- facilitate pronation and supination of the forearm
- provide for rotation of the wrist and hand independent of rotation of the ulna or humerus.

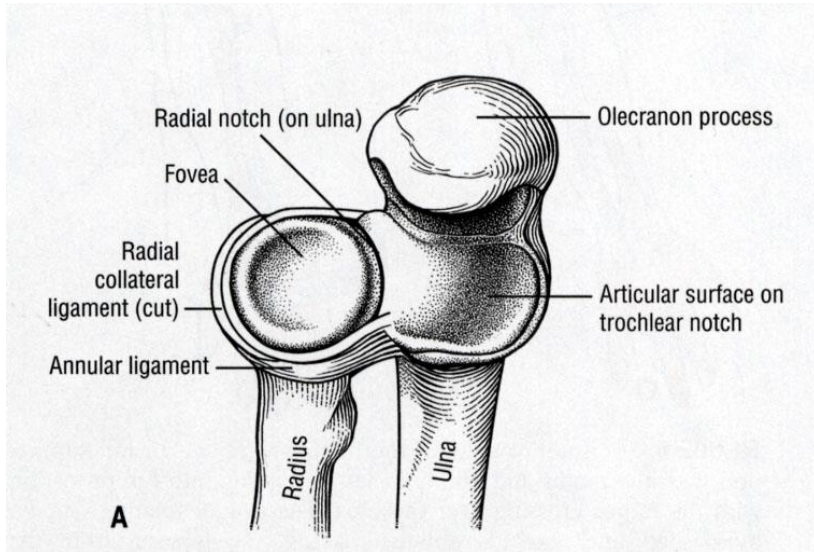
Supination: In anatomic position, the radius and ulna are positioned in parallel to each other thus placing the forearm in full supination.

Pronation: The radius (together with the wrist/hand) rotate & pivot about a relatively fixed ulna such that the radius and ulna are no longer parallel to each other.

There are **three** radioulnar joints:
2 synovial joints and one fibrous joint

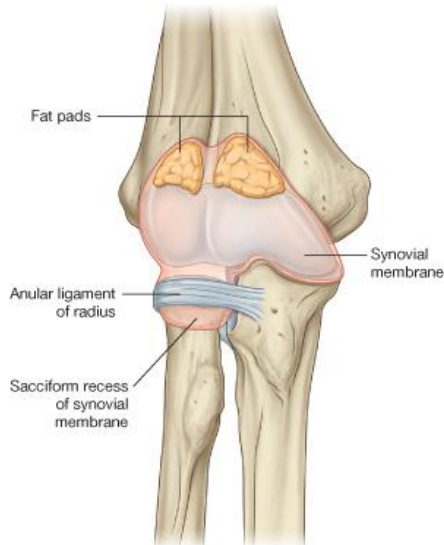
- **Proximal radioulnar joint:** a synovial joint between the **radial head**, **annular ligament** and the **radial notch of the ulna**
- **Middle radioulnar joint:** a fibrous joint comprising an **interosseous membrane** attached to the **crests** of the radius and ulna
- **Distal radioulnar joint:** a synovial joint between the **ulnar head** and the **ulnar notch of the radius**





The proximal radioulnar joint is a synovial joint comprised of:

- Radial notch of ulna, which is *lined with articular cartilage*.
- Radial head/rim, which is covered with a synovial membrane.
- Annular ligament, which attaches from the anterior to posterior aspect of the radial notch of the ulna, is lined with articular cartilage, and encircles the radial head/rim.



: Drake et al: Gray's Anatomy for Students - www.studentco

The radial notch of the ulna and annular ligament, both lined with articular cartilage, provide the articular surface for the radial head.

The radial head **spins** in the radial notch of the ulna to allow the shaft of the radius to pivot about the ulna for pronation and supination.

Middle Radioulnar Joint

A fibrous joint formed by the union of the crests of the radius and ulna via the interosseous membrane.

Assists in stabilizing the proximal & distal radioulnar joints during forearm pronation and supination.

Important in transmitting compression forces from a load-bearing radiocarpal joint to the ulna so that compression forces are distributed between both forearm bones.

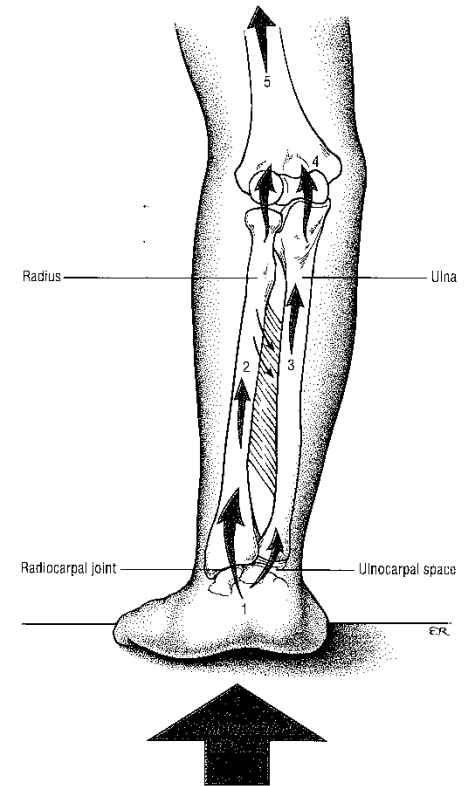
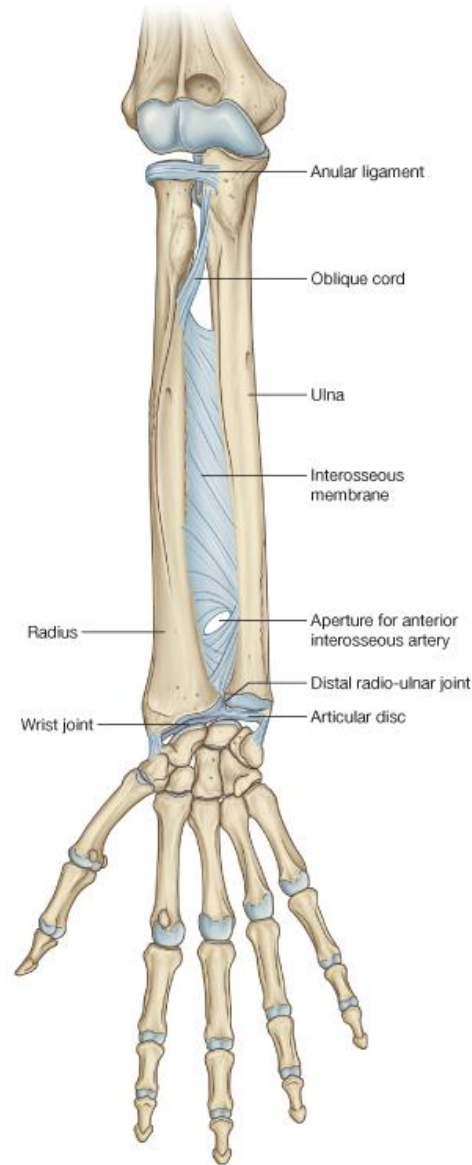


FIGURE 6-20. A compression force through the hand is transmitted primarily through the wrist (#1) at the radiocarpal joint and to the radius (#2). This force stretches the interosseous membrane (shown by double taut arrows) that transfers a part of the compression force to the ulna (#3) and across the elbow at the humeroulnar joint (#4). The compression forces that cross the elbow are finally directed toward the shoulder (#5). The stretched (taut) structures are shown as thin elongated arrows.

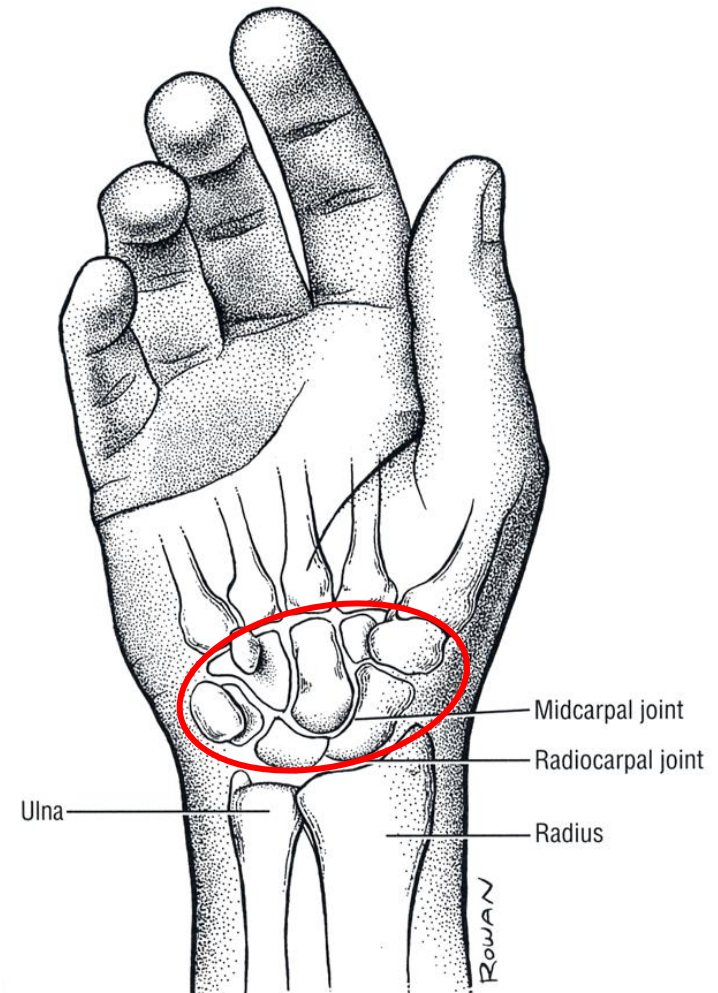
Wrist & Hand Joints

The “wrist” is really a **composite joint** comprising two sets of joints: the radiocarpal and the midcarpal joints. More motion occurs at the radiocarpal joint than the midcarpal joint but full wrist range of motion requires both joints to be able to move normally. Both sets of joints are synovial joints reinforced by ligaments.

The wrist is ellipsoid in shape and provides for 2° of freedom: **flexion and extension and radial and ulnar deviation (or wrist abduction and adduction, respectively).**

The primary function of the wrist joints is to **regulate the amount of muscle-tendon force and range of motion of the extrinsic muscles that control hand and finger movements.**

Consider that the tendons of the long digital flexors and extensors cross many joints and each joint crossed can be acted upon. Each time a joint is acted upon it “uses up” some of the force and ROM that can be produced by a muscle-tendon unit. A multi-joint muscle becomes “actively insufficient” when it reaches a length at which it can no longer effectively generate active force.



For example, flexion force, and the ability to make a full fist (i.e., full active ROM), decrease as the number of joints acted upon by the long digital flexors increases.



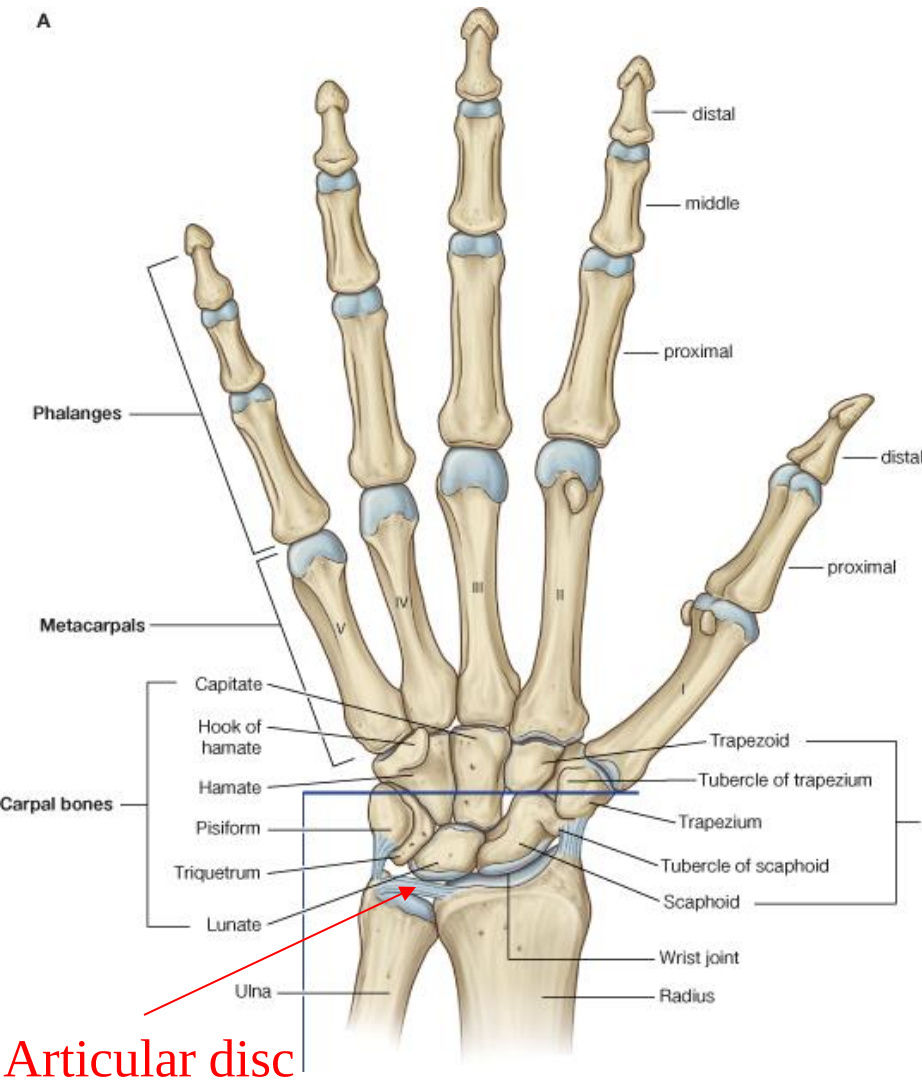
Wrist lock: Disarming technique by grabbing the wrist and forcing the hand into flexion



Radiocarpal Joint

The radiocarpal joint:

- proximal row of carpal bones (scaphoid, lunate, and triquetrum)
- distal radius
- articular disc that attaches from the ulnar notch of the radius to the ulnar styloid process.



This joint is enclosed by a strong, but loose fibrous joint capsule and is reinforced by extrinsic ligaments that limit both radial & ulnar deviation and wrist flexion & extension.